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Magnetic and tectonic history of the Late Proterozoic Upper Little Dal and Coates Lake Groups of northwestern Canada[α]

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ABSTRACT

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A pre- to syn-rift sequence (upper Little Dal Group, Coates Lake Group) in the Mackenzie Mountains of northwestern Canada reveals paleomagnetic directions in pigmental hematite (TH_B, RR_B) that are similar to hematite overprint directions (P2) of underlying Late Proterozoic strata. Paleopoles of these magnetic components from the Thundercloud and overlying Redstone River formations of the Coates Lake Group are respectively 202°W, 27°N ($N=6$ sites, $A_{95}=9°$) and 206°W, 32°N ($N=4$ sites, $A_{95}=10°$).

The primary paleopole for the basal Thundercloud Formation (TH_A) at 210°W, 01°N ($N=6$ sites, $A_{95}=21°$) is close to a previously reported pole (222°W, 2°N) from diabases (dated by U–Pb at 778 ± 2 Ma) that intrude strata below Little Dal Group. The large error in TH_A, caused by scatter along a small circle, is here attributed mainly to tectonic rotation of azimuths of section-mean directions. Based on this interpretation, the TH_A pole position suggests a minimal unconformity below the Thundercloud Formation, contemporaneity of the sills and the basalts that cap the Little Dal Group, and uncomformity between Thundercloud and Redstone River Formations. Anomalous directions from the Little Dal basalts, and sequential directions in magnetic phases of the Redstone River Formation are also interpreted to record local block rotations during trans-tensional tectonism.

The P2 disseminated pigmental hematite magnetization is interpreted to immediately postdate magnetizations from most other hematite phases, as well as magnetizations from (reduced) sulphide/magnetite beds at the base of the Coppercap Formation (upper Coates Lake Group). P2 is absent from, and likely predates deposition of, the upper Coppercap and overlying formations.

Some poles (P3) that postdate P2 poles and indicate high paleolatitudes are here interpreted as Late Precambrian. Negative fold tests suggest that most other high-latitude poles from Late Proterozoic units of the region are of Cretaceous Laramide age.

Introduction

The Mackenzie Mountains in the northern Canadian Cordillera host a well exposed, weakly metamorphosed, and relatively complete Late Proterozoic (<1200 Ma) stratigraphic sequence (Fig. 1), comprising the Mackenzie Mountains Supergroup, the Coates Lake Group, and the Windermere Supergroup, as defined by Jefferson and Parrish (1989). Rifting was incipient during deposition of the upper Little Dal Group, reached a maximum at the base of the Windermere Supergroup, and continued episodically during sedimentation of the Windermere Supergroup. Continental separation (e.g. Eisbacher, 1985) has been inferred at about 550 Ma by Bond and Kominz (1984). Effects of rifting, such as development of unconformities and karst features, diabase intrusions and basalts, tectonically active but confined sedimentary basins, and hydrothermal activity, provided

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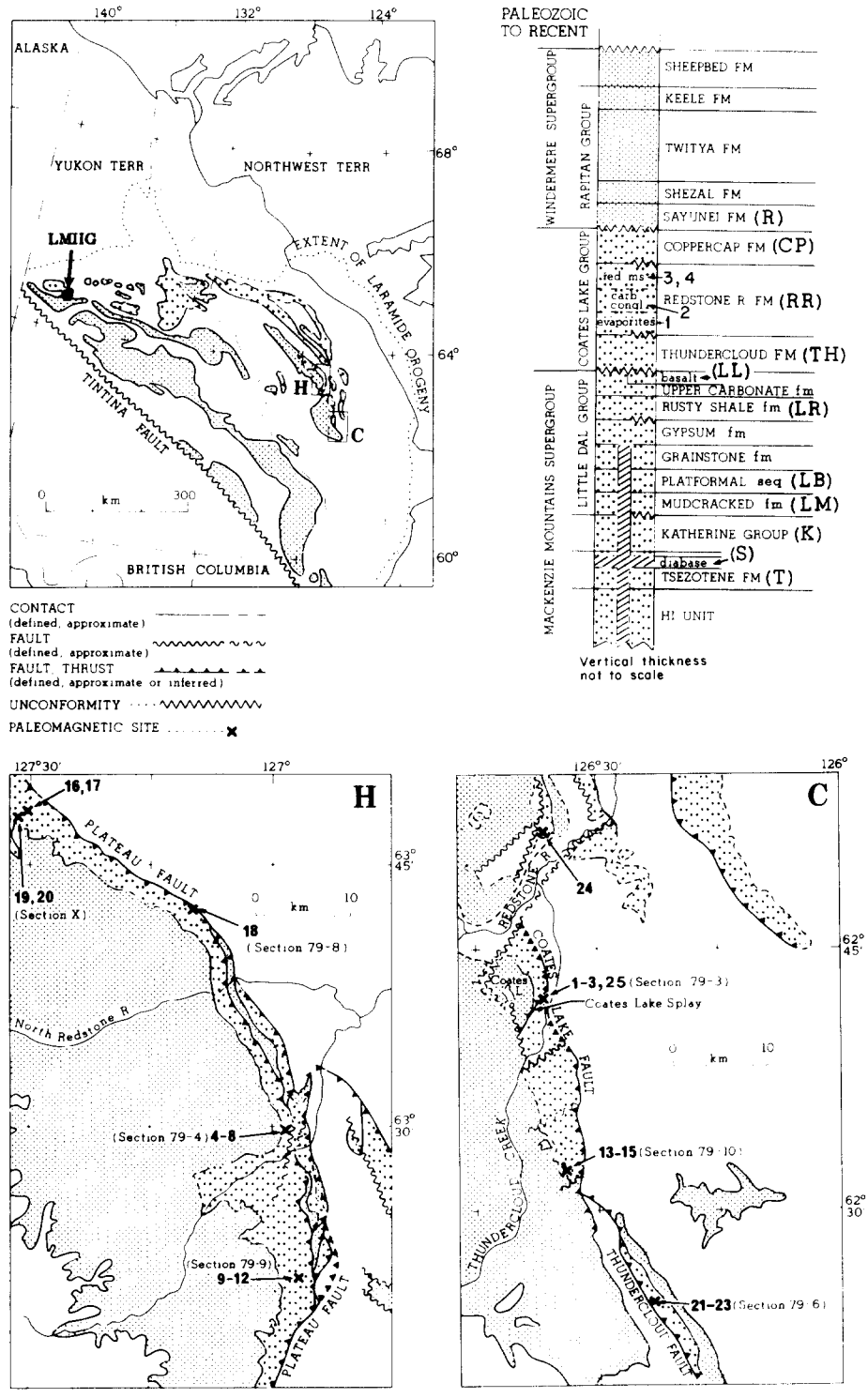


Fig. 1. Location map and regions of study (H, C). H, the Hayhook Lake area, is after Gabrielse et al. (1973) and Jefferson (1983). C, the Coates Lake area, is after Jefferson and Parrish (1989).Lower Mount Harper Group (*LMHG*) is documented by Mustard and Donaldson (1990).

conditions suitable for the sequential formation of stratabound zinc, copper, and iron deposits (Delaney et al., 1982).

Geochronological constraints on the above stratigraphic events are few (Jefferson and Parrish, 1989). In order to improve such constraints, paleomagnetic characterization of these units has been under way for more than a decade (Morris, 1977; Morris and Park, 1981; Park, 1981a, b, 1984; Morris and Aitken, 1982; Park and Aitken, 1986a, b; Park et al., 1989).

Several alternative apparent polar wander (APW) paths have been constructed using primary and secondary magnetic components (Morris, 1977; Park, 1981a; Morris and Park, 1981; Park and Aitken, 1986a, b). Primary poles from the Mackenzie Mountains Supergroup form a reasonably consistent APW path that could join the older Grenville Loop near the Equator in the Pacific Ocean (Park and Aitken, 1986a), but most secondary poles are not easy to relate to this path. One specific problem is whether poles, which indicate high paleolatitudes and are derived from hematite pigment in glaciogenic strata, are Late Precambrian or Cretaceous in age.

This study was initiated to test the significance of the Coates Lake Group's basal unconformity, which was documented stratigraphically by Aitken (1981), paleomagnetically by Morris and Aitken (1982), and further described by Jefferson and Parrish (1989). Results from this study also provide constraints on the nature of rifting, high-latitude poles, and the relative ages of hematite pigments. These results also help to interpret other unconformities such as that between the Redstone River and Thundercloud formations of Coates Lake Group, and that between the Coates Lake Group and the overlying Rapitan Group.

The first author (JKP) is responsible for most of the sampling and all of the paleomagnetic analysis. The second author (CWJ) sampled the basal Coppercap Formation and is responsible for the stratigraphic and structural context of the paper. JKP compiled and interpreted the results, with contributions from CWJ.

Geology

Regional geology

The stratigraphy of Late Proterozoic strata in Mackenzie Mountains is summarized in Fig. 1. The oldest major unit, Mackenzie Mountains Supergroup (see Jefferson and Parrish, 1989 for history of definition) is a platformal sequence of carbonate, evaporite, and generally mature terrigenous detrital strata (Aitken, 1982) which were deposited under warm climatic conditions. Evidence of incipient rifting in the middle of the Little Dal Group is presented by Jefferson and Parrish (1989).

The Mackenzie Mountains Supergroup postdates the Mackenzie igneous events (1270 ± 4 Ma, Le Cheminant and Heaman, 1990). The supergroup's lower part is intruded by diabase earlier dated at $766–769 \pm 27$ Ma by the Rb–Sr method (Armstrong et al., 1982) but recently dated at $777.7^{+2.5}_{-1.8}$ Ma by the U–Pb zircon method (Jefferson and Parrish, 1989) and 779 Ma (1.8% discordant) by $^{207}Pb/^{206}Pb$ on a single baddeleyite fraction (Heaman et al., 1991). It is capped by the Little Dal basalts which are undated but chemically similar to the diabase (ibid.). The Mackenzie Mountains Supergroup is lithostratigraphically correlated with the Shaler and Rae Groups of the Canadian Shield (Young et al., 1979).

The Coates Lake Group, bounded by unconformities, discontinuously separates the underlying Mackenzie Mountains and overlying Windermere supergroups along the Plateau Thrust zone, and hosts the Redstone Copper Belt (Jefferson and Ruelle, 1986). The Coates Lake Group shares a warm climatic setting with the underlying Mackenzie Mountains Supergroup (red beds and evaporites), but shares strong evidence of rift tectonics with the basal

Windermere Supergroup (Jefferson and Parrish, 1989). Aitken (1981) and Gabrielse and Yorath (1989) assign the Coates Lake Group to the Windermere Supergroup. Jefferson and Parrish (1989) and this paper separate the two, because the Coates Lake Group's mapping characteristics and metallogeny are more similar to the Mackenzie Mountains Supergroup.

The Windermere Supergroup differs from the underlying Mackenzie Mountains Supergroup and Coates Lake Group by its basinal setting, mainly drab colour, and series of shoaling-upward grand cycles (Aitken, 1989). Its heterogeneous clastic strata in many areas record deposition under cold climatic conditions (Young et al., 1979; Eisbacher, 1981). Dramatic syn-sedimentary faulting related to rifting controlled the facies architecture of basal units such as the Sayunei Formation (Eisbacher, 1981) and the Lower Mount Harper Group (Mustard and Donaldson, 1990), which is located some 500 km northwest of the study area (Fig. 1). The most dramatic unconformity and the greatest fault offsets in the Mackenzie Mountains region are at the base of the Rapitan Group (Jefferson and Parrish, 1989). The Windermere Supergroup extends the length of the Cordillera but has no correlatives on the Canadian Shield (Young et al., 1979).

Some folds within the Sayunei Formation may be related to block faulting and compression before deposition of the Shezal Formation (Helmstaedt et al., 1979; Jefferson and Parrish, 1989). Aitken et al. (1981) have disputed such interpretations. The great majority of mapped folds are probably Cretaceous in age (Gabrielse et al., 1973), associated with thrust faults and weak metamorphism (Hofmann and Aitken, 1979).

Stratigraphy of Coates Lake Group

The Coates Lake Group (Jefferson and Ruelle, 1986) consists of three formations: the Thundercloud Formation (Jefferson, 1983) (basal sandy member of Aitken, 1981), the Redstone River Formation (Gabrielse et al., 1973), and the Coppercap Formation (ibid.) (Fig. 1).

The basal Thundercloud Formation comprises carbonate, clastic, and evaporitic rocks, recording shallow marine transgression upon alluvial orthoconglomerates that were derived mainly from the Little Dal basalts. The upper Thundercloud Formation is characterized by rhythmically interbedded sabkha dolostones and maroon mudstones.

The Restone River Formation overlies the Thundercloud gradationally in basinal areas (informal number RR1 = marine/lacustrine evaporites), and abruptly in marginal areas (RR2 = thin carbonate conglomerates). Thick sequences of evaporitic, hematitic, alluvial siltstone rhythmites (RR3, 4) constitute the bulk of the Redstone River Formation, and are interbedded with the underlying evaporites, laterally equivalent conglomerates, and the overlying Coppercap Formation. The RR2 conglomerate and paleomagnetic results presented in this paper suggest, however, that an unconformity is present between the RR3 maroon siltstones and the underlying Thundercloud Formation.

The basal member of the Coppercap Formation (CP1, also known as "Transition Zone") records a sabkha environment; it consists of evaporitic, microbial carbonates interdigitated with the maroon siltstone rhythmites. The remaining strata of the Coppercap Formation comprise two shoaling-upward sequences of predominantly carbonate turbidites capped by stromatolites. The CP1 carbonates host undeveloped yet significant stratabound and stratiform copper deposits (Ruelle, 1982; Jefferson and Ruelle, 1986). Petrographically similar but minor copper occurrences are located at several other stratigraphic levels in the Coates Lake Group, at the base of the Rapitan Group (below glaciogenic strata), and in the underlying Mackenzie Mountains Supergroup (ibid.).

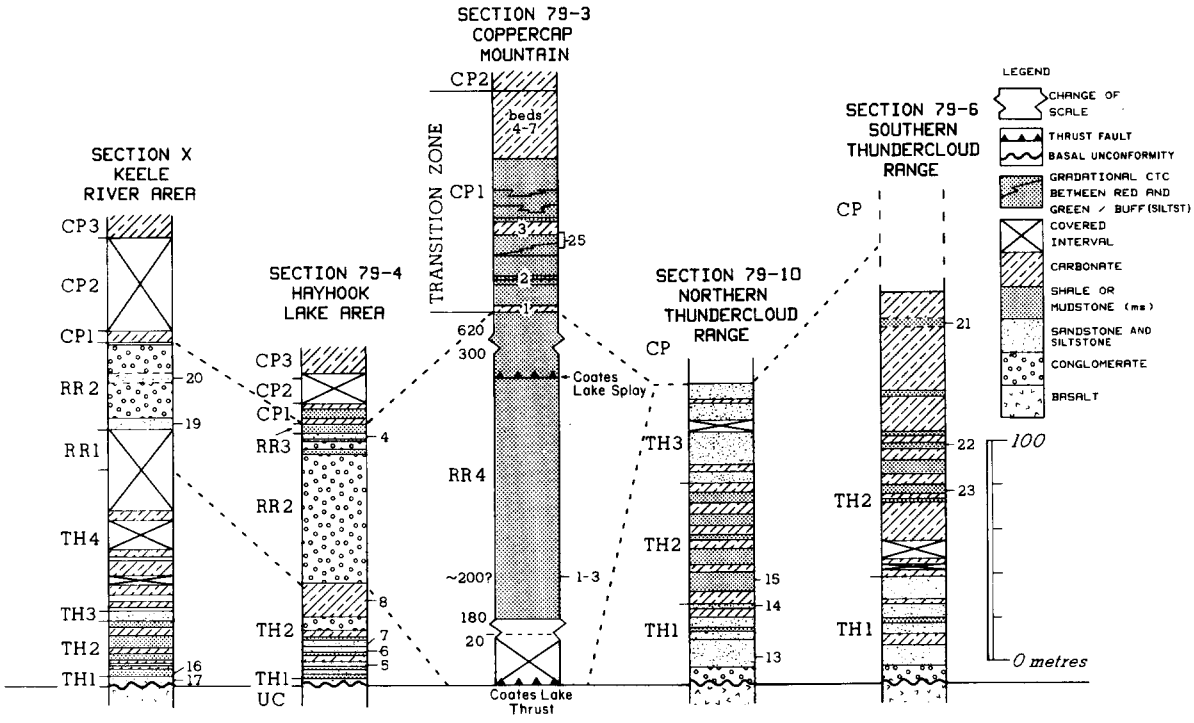


Fig. 2. Stratigraphic sections sampled from the Coates Lake Group. Sections 79-3, 79-4, 79-6, 79-10 refer to sections of Aitken (1981). Sections 79-3 and 79-4 are equivalent to sections 12 and 49 of Jefferson (1983), and section X is after section 56 (Jefferson, 1983), located nearby. Sites are plotted, approximately, on the right side of columns. The members (TH1, RR1, etc.) are as defined by Jefferson and Ruelle (1986).

Sample locations

Samples for paleomagnetic study were collected from the Rusty Shale formation (informal term; Aitken, 1981) (5 sites) and basalt (1 site) of the upper Little Dal Group, and from the Thundercloud (12 sites), Redstone River (6 sites), and basal Coppercap formations (1 site) of the Coates Lake Group (Fig. 1). Generalized stratigraphy of the geological sections and the approximate stratigraphic locations of sites are given in Fig. 2. Most sections dip to the west with shallow (about 10°; X, 79-9), moderate (40–45°; 79-3, 79-4, 79-8), or steep (50–60°; 79-10) inclinations; one section (79-6) dips steeply (65°) to the northeast (see Table 7 for bedding corrections).

Methods

Sampling

Hand samples or cores were collected with helicopter support in 1981 and 1986. Samples were oriented in the field by sun and magnetic compasses. In the laboratory, cores were drilled from hand samples and cut into specimens of standard size (2.2 cm long, 2.5 cm diameter).

Demagnetization experiments

The natural remanent magnetization (NRM) of specimens was measured, and a number of experimental runs were conducted over several years, involving alternating field (AF), thermal (TH), and chemical (CH)

techniques to resolve the NRM. Some techniques were combined, with TH following AF treatment, or AF and (or) TH following CH treatment. Thermal demagnetization in air was generally done on an "in-house" apparatus (Roy et al., 1972) with some use of a commercial instrument (Schonstedt TSM-1). Also an in-house instrument (Roy et al., 1973) was mainly used for AF studies. Residual fields in the in-house instruments were generally less than 2 nT, but ranged up to 30 nT in the commercial TSM-1. After each step of TH or AF treatment, remanent magnetizations were measured on spinner magnetometers, either a Schonstedt DSM-1 or a Geofyzika JR-4. During each run, samples were stored in commercial magnetic shields.

Acid leaching

Acid leaching, involving the physical separation of magnetic phases, utilized the following method. First, an axial hole (1 cm diameter) was drilled in specimens to provide a larger surface area for acid access and egress (Park, 1970). Specimens (183 in total) were then immersed in HCl of varying strength (6–10 *N*), depending on their carbonate content and their resistance to leaching, for periods of 8 hours to 28 days. Many specimens (about 20%) disintegrated because of the carbonates present, but others were leached up to 4000 hours over a period of a year. After each stage of leaching, specimens were immersed in water for several days, rinsed, dried, put under vacuum to evacuate as much of the remaining acid as possible, and allowed to dry further. Leaching was conducted under vacuum (Henry, 1979; Roy and Anderson, 1981), which allowed faster acid penetration, reducing leaching times by a factor of two (Park and Aitken, 1986a, b). All leaching was done within an environment of reduced magnetic field in order to inhibit the acquisition of a VRM (viscous remanent magnetization) (Roy and Park, 1974) and (or) CRM (chemical remanent magnetization) (Henry, 1979). Some unwanted magnetic components were detected and nullified by routinely inverting specimens on alternate treatment steps (Park and Aitken, 1986a, b). In addition, after drying between steps, but before being measured, specimens were placed in portable magnetic shields for periods equal to the leaching time of the last stage in order to allow any acquired soft VRM to decay (Roy and Park, 1974).

When leaching was essentially complete, as judged by resistance to further decay of magnetic intensity and confirmed by slicing some representative specimens to ensure that the pigment had been largely removed, some specimens were subjected to AF and (or) TH treatment. This further treatment indicated (1) whether any CRM or VRM acquired during the experiment remained and (2) the magnetic characteristics of the remanent magnetization.

Determination of magnetic components

Magnetic components reported in this paper generally have statistically distinct directions, though they may comprise several different or related coeval magnetic phases. For example, a primary component may reside in detrital magnetite and hematite, and a secondary component may reside in diagenetic hematite as a disseminated phase and as coatings on quartz. In the Redstone River Formation, components RR_F and RR_B have overlapping directions, but directions and phases are considered to be sufficiently different to justify distinguishing them as different components. Magnetic phases may be distinguished by their magnetic characteristics, by their order of removal during demagnetization, and by microscopic observations. These distinguishing aspects not only served to relate phases to magnetic components but also to determine the relative timing of components.

Results from CH and attendant TH demagnetization studies of Henry (1979) and Park

and Aitken (1986a, b) served as models for defining some components. Henry (1979) found that fine-grained hematite pigment was most soluble in acid. It was removed while specularite (detrital hematite) was only partially leached. Titanomagnetite was rather insoluble. He found that several phases of hematite, similar to ones occurring in this study, were removed by acid in the same order that they were thermally demagnetized. Thus, a later stage of hematite pigment was demagnetized from 275 to 450°C, an earlier stage of hematite pigment was demagnetized from 450 to 625°C, and the specularite was demagnetized from 625 to 665°C. Park and Aitken (1986a, b) also found that hematite pigment had low unblocking temperatures, often less than 550°C. The CH technique was crucial to the interpretation of components. It selectively removed phases, so that characteristics of removed and remaining phases could be determined.

Magnetic components were mainly determined from vector diagrams (Zijderveld, 1967; Roy and Park, 1974) by visually fitting straight lines through apparent straight-line segments. When the LINEFIND computer program (Kent et al., 1983) became available, results from several sites were checked and corroborated. The visual method is not as accurate, but can be as objective as computer methods (e.g. Kirshvink, 1980; Schmidt, 1982; Kent et al., 1983) for analyzing vector data, considering subjective error parameters assigned by the observer. Any observer biases will tend to be removed when large numbers of specimens are processed, as in this study.

Statistical analysis of results was complicated by the following: (1) the many different

TABLE 1
Mean site and component directions from Rusty Shale formation

Comp.	Si.	Sa.	Sp.	Det.	Treatm.	$D°, I°$ (w.r.t. hor.)	$D°, I°$ (w.r.t. bdg.)	R	k	$\alpha_{95}°$	Pol.
LR_A	9	5	17	19	1, 4, 7	286, +11	286, −13	16.51	32	3	N
	11	3	5	5	1, 6, 7	095, +01	094, +26	4.70	13	22	R
	12	3	9	9	1, 6, 7	088, +05	085, +27	8.88	68	6	R
	Mean of three sites					*277, +03*	–	*2.96*	*49*	*18*	*M*
						–	*277, +21*	*2.96*	*51*	*17*	*M*
LR_{C1}	9	5	12	12	2, 4, 6	021, +86	300, +66	11.89	100	4	N
	10[a]	5	10	13	1, 2, 4	012, +63	337, +53	9.03	9	17	N
	11	4	13	16	1, 2, 4, 6, 7	089, +67	347, +83	12.55	27	8	N
	12	3	6	8	1, 4, 7	073, +78	320, +74	5.92	60	9	N
	Mean of four sites					*051, +76*	–	*3.90*	*29*	*17*	*N*
						–	*324, +70*	*3.90*	*31*	*17*	*N*
LR_{C2}	10	5	18	18	1, 6, 7	060, +54	024, +63	16.88	9	12	R
	11	4	8	8	1, 7	087, +29	078, +53	7.75	28	11	R
	12	3	7	7	1, 7	082, +37	067, +57	6.73	22	13	R
	Mean of three sites					*078, +40*	–	*2.92*	*25*	*25*	*R*
						–	*059, +60*	*2.93*	*28*	*24*	*R*
LR_{C3}	18	5	10	10	1, 5	354, +86	233, +60	9.81	47	7	N

Comp. (components). Si. is site number; Sa., Sp., and Det. are the number of samples, specimens, and determinations, respectively. Several determinations of the same direction were made on some specimens using different treatments. These determinations may document the direction in one or several magnetic phases. In some cases, the same treatment revealed the same direction in several magnetic phases. The treatment types are indicated in Figs. 4, 6, and 11: for example, treatment type 6 indicates that a sample has undergone AF and TH treatments to uncover the component recorded. $D°, I°$ are the declination and inclination of the direction. R is the vector resultant. k is Fisher's (1953) precision parameter. $\alpha_{95}°$ is the error in direction, the half angle of 95% confidence. Pol. is polarity, where N is normal (directions with a westerly component of dip), R is reverse, and M is mixed. Site statistics determined using specimen unit weight. Mean component directions are in italics.

[a] Excluding anomalous values from one sample of site 10.

TABLE 2
Mean site and component directions from Thundercloud Formation

Comp.	Sect.	Si.	Sa.	Sp.	Det.	Treatm.	$D°, I°$ (w.r.t. hor.)	$D°, I°$ (w.r.t. bdg.)	R	k	$\alpha_{95}°$	Pol.
TH_A	79-4	6	1	1	1	6	290, +52	269, +05	–	–	–	N
		7	2	4	4	7	297, +57	282, +23	3.75	12	28	N
	X	8	2	2	2	6, 7	309, +36	297, +09	1.88	–	–	N
		16	2	2	2	7	252, +10	251, +04	1.99	–	–	M
		17	1	5	5	4, 6, 7	262, +17	260, +11	4.84	25	16	N
	79-6	21	2	2	2	4	044, +76	050, +10	1.98	–	–	R
		23	5	5	12	1–4, 6	272, −54	258, +01	4.86	28	15	M
	79-10	13	4	10	12	1, 5, 7	298, +60	268, +10	8.72	7	20	M
	Mean of six[a] sites						*(279, +12)*	–	*3.36*	*2*	*69*	*M*
							–	*264, +06*	*5.51*	*10*	*22*	*M*
	Mean of 31 specimens						*(278, +26)*	–	*19.09*	*3*	*21*	*M*
							–	*264, +09*	*27.80*	*9*	*9*	*M*
TH_B	79-4	5	4	10	16	1, 4–7	335, +79	275, +49	9.79	44	7	N
		6	5	19	27	1, 2, 4–7	018, +83	275, +53	18.69	58	4	N
		7	5	9	10	5, 6, 7	358, +80	280, +52	8.90	78	6	N
		8	3	5	6	6, 7	006, +80	273, +54	4.92	51	11	N
	Mean of four sites						*358, +81*	–	*4.00*	*606*	*4*	*N*
							–	*276, +52*	*4.00*	*851*	*3*	*N*
	X	16	2	4	5	6, 7	280, +53	269, +49	3.90	31	17	N
		17	5	9	9	1, 7	(201, +73)	(204, +63)	6.60	3	33	N
	79-6	23	3	3	3	7	248, −24	250, +35	2.98	84	14	N
	79-10[b]	13	4	4	5	3, 4	298, +86	248, +27	3.99	289	5	N
		14	6	15	21	1, 2, 4–7	306, +72	277, +21	14.83	84	4	N
		15	4	16	25	1, 3–7	298, +82	268, +35	15.90	146	3	N
	Mean of three sites, 13–15						*303, +80*	–	*2.98*	*125*	*11*	*N*
							–	*265, +28*	*2.93*	*29*	*23*	*N*
	Mean of nine sites, excluding site 17						*290, +74*	–	*7.56*	*6*	*24*	*N*
							–	*268, +42*	*8.72*	*35*	*13*	*N*
	Mean of sites 5–8, 16, 23						*(285, +71)*	–	*4.60*	*4*	*41*	*N*
							–	*269, +49*	*5.92*	*63*	*9*	N
TH_C	79-6	21	5	12	12	1, 6, 7	248, +39	013, +71	11.81	57	6	N
		22	5	8	8	3, 6, 7	257, +47	040, +66	7.86	49	8	N
	Mean of two sites						*252, +43*	–	*1.99*	*123*	*23*	*N*
							–	*028, +69*	*1.99*	*109*	*24*	*N*

See Table 1 for explanation of symbols. Data in parentheses, with precisions less than 5, are arbitrarily rejected. Site statistics determined using specimen unit weight.
[a] Excluding one-sample sites.
[b] Data may reflect unknown tilting.

treatments used, sometimes two or three on the same specimens; (2) the fact that several specimens (as many as five) were treated from the same sample; and (3) the fact that the same treatments were not always effective on companion specimens. Because several different magnetic directions were found in the samples, and there was apparent overlap among the directions of the magnetic phases, directions were exhaustively compared in terms of treatments, treatment ranges, specimens, sites, and geologic sections, and the petrography of specimens, before assignment of directions to specific components was finalized. Different treatments in some cases determined the same magnetic component; therefore, similar components revealed by different treatments in the same specimen are combined. Some, if not

TABLE 3
Mean site and component directions from Redstone River Formation

Comp.	Sect.	Si.	Sa.	Sp.	Det.	Treatm.	$D°, I°$ (w.r.t. hor.)	$D°, I°$ (w.r.t. bdg.)	R	k	$\alpha_{95}°$	Pol.
RR_F	79-3	1	5	18	18	1, 6, 7	294, +78	268, +35	17.83	100	3	N
		2	6	30	30	1, 6, 7	354, +84	273, +52	29.55	64	3	N
		3	4	12	12	1, 6, 7	333, +85	277, +47	11.89	98	4	N
	Mean of three sites						*318, +83*	–	*2.99*	*210*	*9*	*N*
							–	*272, +45*	*2.97*	*75*	*14*	*N*
	79-4	4	3	4	4	7	000, +82	282, +52	3.96	76	10	N
	X	19	4	5	5	6	279, +35	274, +29	4.86	28	14	N
		20	5	7	7	1, 5	269, +30	266, +23	6.85	41	10	N
	Mean of six sites					*1, 5–7*	*286, +67*	–	*5.43*	*9*	*24*	*N*
							–	*273, +40*	*5.87*	*38*	*11*	*N*
RR_B	79-3	1	5	14	15	1, 2, 5–7	333, +80	273, +42	13.85	86	4	N
		2	6	21	25	1, 2, 5–7	359, +81	277, +52	20.36	31	6	N
		3	4	11	12	1, 2, 5, 7	010, +84	280, +51	10.95	183	3	N
	Mean of three sites						*351, +82*	–	*3.00*	*562*	*5*	*N*
							–	*276, +48*	*2.99*	*183*	*9*	*N*
	79-4	4	2	3	3	5–7	061, +78	280, +62	2.99	195	9	N
	X[a]	19	3	6	6	6, 7	109, −31	105, −26	5.79	24	14	R
		20	2	4	4	1, 7	309, +77	278, +73	3.96	69	11	N
			2	3	3	3, 4, 6	150, −80	100, −78	2.86	14	34	R
	Mean of four sites (1–4)					*all*	*013, +82*	–	*3.98*	*153*	*7*	*M*
							–	*277, +52*	*3.97*	*92*	*10*	*M*
RR_D	79-3	1	4	4	4	4	356, +61	298, +42	3.95	62	12	N
		2	4	6	6	4	017, +65	306, +55	5.93	75	8	N
		3	1	2	2	4	345, +62	305, +37	1.99	–	–	N
	Mean of three sites						*359, +63*	–	*2.98*	*115*	*12*	*N*
							–	*303, +45*	*2.97*	*67*	*15*	*N*
	79-4	4	1	1	2	1, 4	026, +63	315, +57	1.98	–	–	N
RR_E	79-3	1	5	6	6	3, 4	002, +41	324, +36	5.93	75	8	N
		2	4	9	9	1, 3, 4	008, +51	323, +46	8.95	155	4	N
		3	3	4	4	3, 4	350, +44	322, +28	3.93	40	15	N
	Mean of three sites						*000, +46*	–	*2.98*	*99*	*12*	*N*
							–	*323, +37*	*2.98*	*86*	*13*	*N*

See Table 1 for explanation of symbols. Site statistics determined using specimen unit weight. [a] These directions are unclassified; they could relate to RR_B (see text).

most components are chemical in nature, and on a specimen basis may have been acquired at different times. For this reason, and because similar components in companion specimens may be unresolved, specimens are given unit weight in within-site statistical calculations (Tables 1–3).

Fold test

Statistics of grouped data were compared before and after directions were corrected for bedding by the fold test of McElhinny (1964) to determine whether magnetic components were acquired before or after folding. Recent fold tests proposed by McFadden and Jones (1981) and McFadden (1990) could not be used effectively in the present study. Groups of sites did not occur on the opposite limbs of one fold as required by the McFadden and Jones (1981) test, and the number of limbs on which sites did occur was for all analyses less than the five recommended for use of the more flexible McFadden (1990) test. McElhinny's (1964) test, however, is not strictly valid (McFadden and Jones, 1981). If it is positive,

it will confirm a positive fold test that is indicated by the more recent methods. But if the fold test is negative, it may still prove positive by the other tests. It is therefore likely that the positive tests indicated are more significant than shown.

Designation of components

Capital letters refer to formation or rock unit name and subscripts to a specific magnetic component within the unit. Formations of the Little Dal Group begin with 'L' followed by a letter designating the formation, such as 'R', for Rusty Shale formation. The subscript *generally* indicates the order in which the specific magnetic component was acquired, with 'A' representing the primary or original magnetization. 'B' subscripts mostly refer to a single hematite pigmentation event, P2, and 'C' subscripts to later pigmentation events with high-latitude poles that may or may not be of the same age.

Results and analysis

Petrography of Rusty Shale formation (informal term; Aitken, 1981)

Petrography and magnetic mineralogy of the five Rusty Shale sites (Fig. 1) are variable. Site 9 is a fine-grained purple shale; sites 10–12 and 18 are medium-grained varicoloured sandstones. The magnetic mineralogy of sites 9–12 (geologic section 79-9) is complex, and magnetic phases are difficult to identify. They are described in sequence in the petrographic and following paleomagnetic section.

Detrital hematite is abundant (site 9) to rare (site 11) and contains lenses of ferrian rutile or rutile. Some hematite (martite) may have formed diagenetically from detrital magnetite.

Thin hematite coatings on detrital quartz grains contain another magnetic phase in sites 10–12 and are clearly preserved because of epitaxial quartz overgrowths.

Acicular/platy crystals of hematite occur as a second hematite overgrowth and in a disseminated form. They are especially evident at site 12. Crystals, oriented normal to quartz boundaries, postdate the hematite rims and mostly predate the quartz overgrowths. This phase may be coincident with pervasive red hematite pigment that occurs in sites 10 and 11.

Hematite pigment varies widely in concentration and distribution; in site 18 (section 79-8) it is confined to fractures that are possibly Recent in age.

Euhedral hematite (martite) inclusions after diagenetic magnetite constitute an additional magnetic phase in quartz overgrowths.

Acid leaching confirmed that euhedral hematite, hematite coatings, and some hematite pigment were protected from acid by the quartz overgrowths. The acid, while removing iron, leaves titanium, thus changing "hematite" (titanohematite) to rutile. Other opaque minerals include large detrital grains of pyrite (up to 0.8 mm diameter) (site 10), some of which are altered to goethite; small post-depositional grains of probable chalcopyrite (site 18); and common rutile. Detrital, acicular, and pigmental hematite are altered to rutile in varying degrees; the alteration ranges from rare (site 10), through minor (sites 11, 12), to abundant (site 9).

Magnetic components of Rusty Shale formation

The magnetic opaques, mainly hematite, yield NRMs directed easterly and westerly with shallow to moderate inclinations (Fig. 3a). These magnetizations were subsequently resolved into four magnetic components (LR_A, LR_{C1}, LR_{C2}, LR_{C3}) by several different treatments. Directions of these components are given in Table 1. The ranges of the treatments over which components were revealed and the interpreted magnetic phases hosting them are

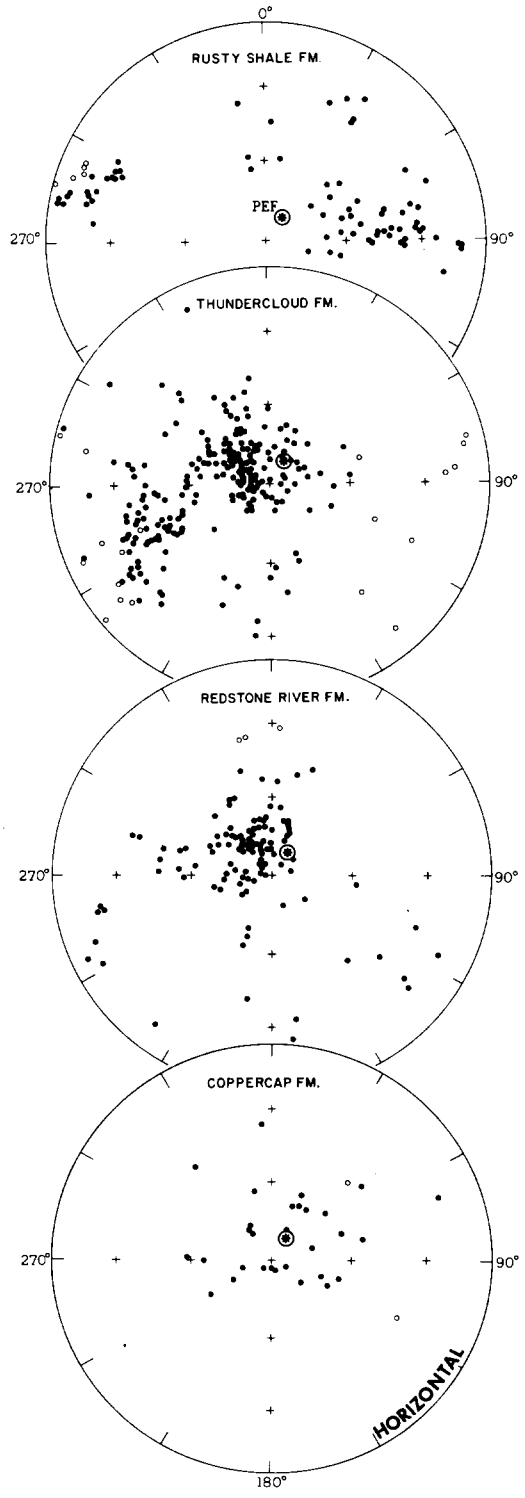


Fig. 3. Natural remanent magnetization of specimens. Closed symbols represent downward-directed magnetizations; open symbols, upward-directed magnetizations. PEF is the direction of the present Earth's field in the region.

shown in Fig. 4. Illustrations of components revealed by different techniques are given in Fig. 5. Magnetizations were almost completely removed between 625 and 650°C. Some measurements are within the noise limit of the instrument (Fig. 5e) or indicate the acquisition of a rotational remanent magnetization (Fig. 5d).

Magnetic components reside in "detrital" (LR_A) and diagenetic hematite phases (LR_{C1}, LR_{C2}, LR_{C3}). The host of LR_A is martite or detrital hematite. LR_A appears to be primary because a reversal of magnetization is recorded between sites 9 and 11 (Table 1).

Component LR_{C1} resides in the first phase of diagenetic hematite, which is disseminated and thinly coats quartz grains. A second phase of diagenetic hematite with an acicular or platy habit contains component LR_{C2} and partly overgrows the first-phase hematite coatings. Partway through formation of the second phase, diagenetic quartz began to grow epitaxially over original quartz grains. Because of its affinity for pre-existing quartz, the quartz overgrowth surrounded first-phase hematite coatings and part of the second phase but left the first phase disseminated hematite largely unenclosed. The acicular phase apparently continued to grow in a disseminated form after final crystallization of the diagenetic quartz. Crystals of the first-phase disseminated hematite may have formed the nuclei on which the second phase grew. A later hematite pigment phase that hosts LR_{C3} occupies fractures of possible Recent or Laramide origin.

Petrography of Thundercloud Formation

Sites in the Thundercloud Formation are summarized in stratigraphic context in Fig. 2. The samples tested are mainly maroon, quartzose, medium-grained sandstones from members TH1 and TH2. The magnetic mineralogy is exceedingly complex. Individual magnetic minerals and their associated non-magnetic

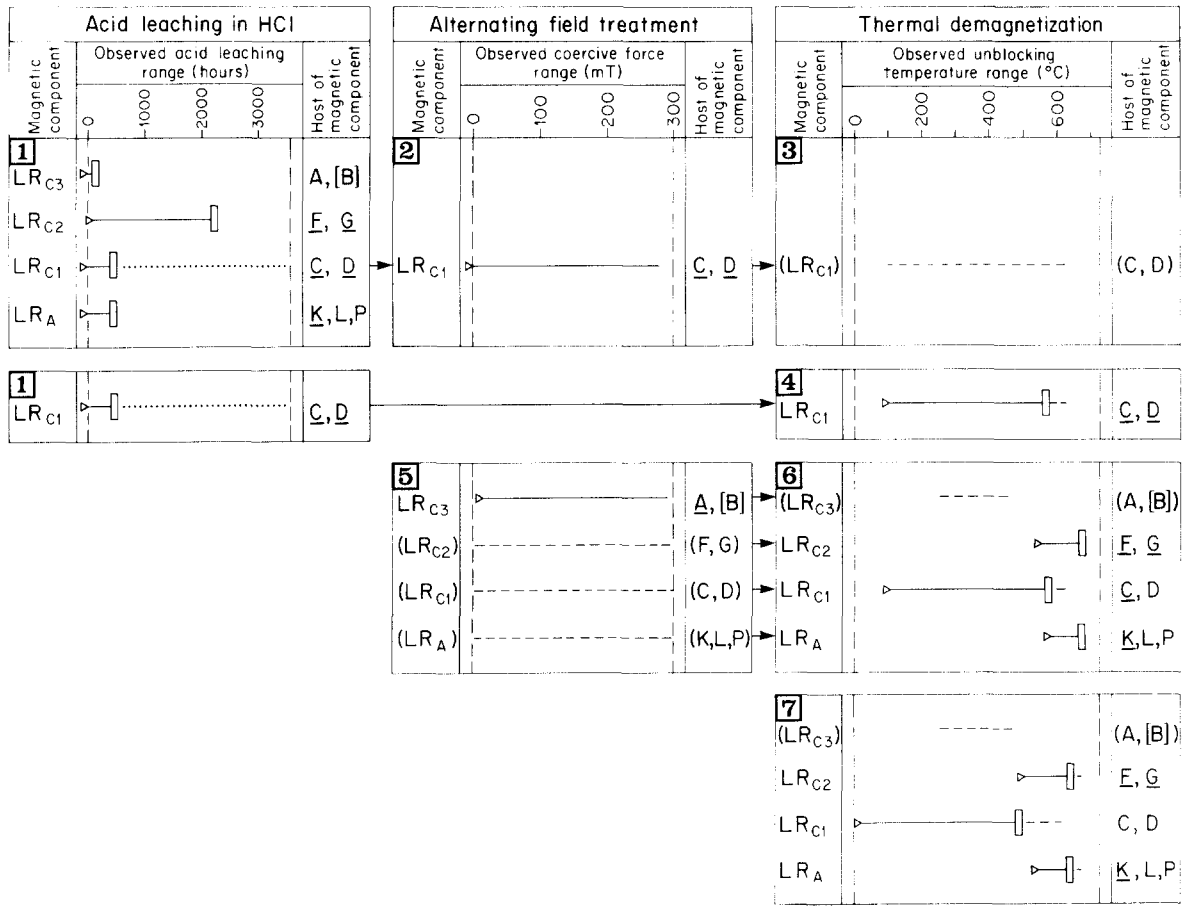


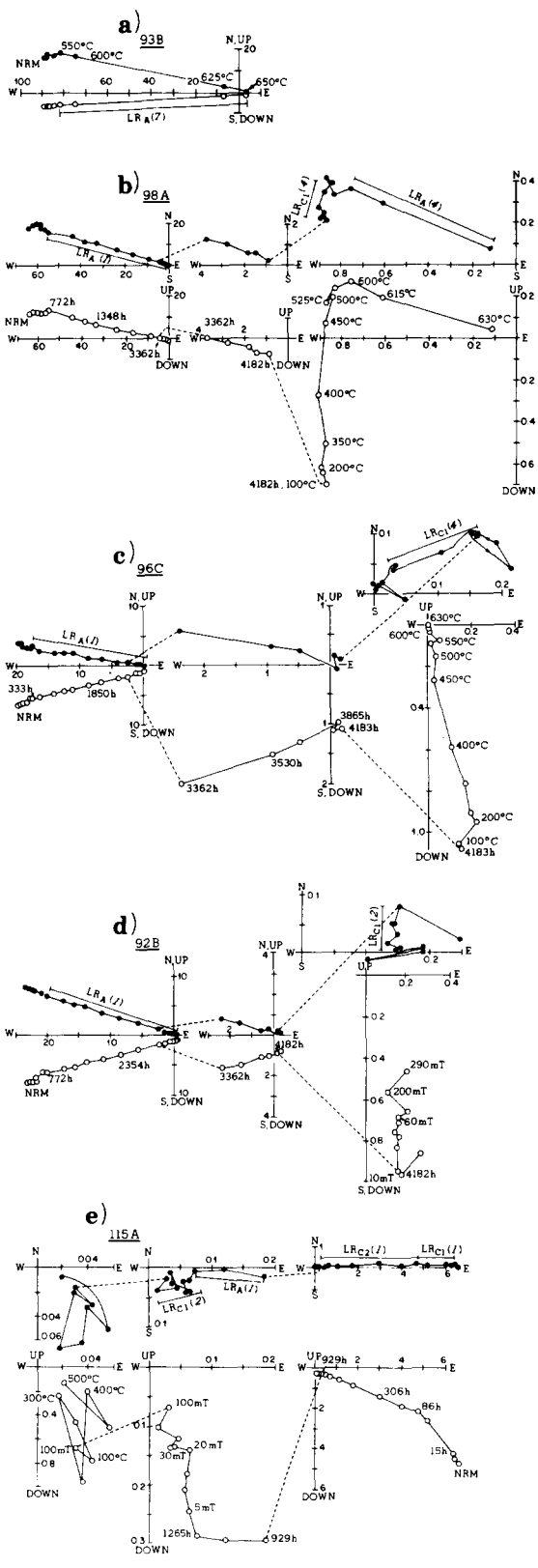
Fig. 4. Documentation of the characteristics of magnetic components from the Rusty Shale formation and the assignment of components to magnetic phases present. Legend in Fig. 6. Magnetic components explained in the Methods section.

opaque phases are difficult to identify with the microscope and the microprobe.

Detrital hematite or *martite* is both sparsely dispersed and (or) concentrated in heavy-mineral laminae (20 to 40% by volume, site 5). Backscattering data indicate that the hematite is finely granulated on the smallest scale and comprises several coexisting opaques. Microprobe determinations revealed rutile and titanohematite, with a TiO_2 content that was measured as high as 9.3%. Microscopic inspection indicates patches or lenses of possible ferrian rutile or rutile coexisting with the "hematite". Where alteration is more advanced, some original hematite grains contain rutile lamellae (sites 16, 17), rutile mantles (sites 13, 15), and rutile pseudomorphs (sites 13–15, 21–23).

Detrital magnetite was petrographically identified only at site 13. Magnetite constitutes up to 100% of some grains. Hematite, except for some extensive rutile alterations, constitutes the remaining parts of magnetite grains as lamellae and as irregular (alteration) rims and invasions along lamellae. About 10% of all framework grains are magnetite or have magnetite in them at this site.

Euhedral grains of magnetite or martite, which occur at site 13 and possibly 21, and hematite at other sites, may be detrital or diagenetic in origin; but they clearly predate quartz overgrowths. Some of these grains are sub-



rounded, suggesting a detrital origin; others are euhedral or irregularly shaped, and clearly diagenetic.

Diagenetic hematite overgrowths of lath-shaped or botryoidal habit are in optical continuity with host opaque grains. Some lath-shaped hematite overgrowths occur at quartz grain boundaries. In site 13, spectacular (thick) hematite overgrowths, probably including goethite, form relatively smooth rims in optical continuity with host opaque grains and may be predepositional.

Minor amounts of colloform and clotted goethite are present as void-fillers.

Thin hematite coatings are common on quartz grains, and are clearly defined and preserved by epitaxial quartz overgrowths. In site 21, two layers of hematite separated by silica are clearly evident, surrounding one grain. Thicker coatings on the rare quartz grains at some sites are remarkably continuous and may, like those on opaque hosts, have a predepositional origin.

The matrix hematite pigment is abundant at most sites. In some samples this pigment is separate from hematite that coats grains, in others the two are juxtaposed. At site 5, early diagenetic epitaxial quartz overgrowths separate hematite coatings from the matrix hematite pigment, which is cemented by a later quartz phase. At site 6, some hematite pigment is located within carbonate cement, which was mainly introduced later than the quartz.

Disseminated hematite pigment includes three petrographically distinct phases: (1) disseminated hematite (too fine-grained to re-

Fig. 5. Vector diagrams (Zijderveld, 1967) of the demagnetization of various specimens from the Rusty Shale formation using different treatments or combinations of treatments. Plots with respect to the horizontal (N–E, solid dots) and vertical (UP-E, open circles) are shown separately or superimposed. Dashed lines link identical points where scales have been expanded. All vectors are depicted relative to the present horizontal. Magnetic components removed are indicated, together with a code representing the treatment (in parentheses; see Table 1, Fig. 4). All scales are in mA/m.

solve optically) located between framework grains; (2) spheroidal clusters of radiating acicular hematite and rutile preserved in quartz overgrowths; and (3) patches of hematite within relict mineral/rock fragment grains, which have been completely altered to sericite ± chlorite ± iron oxides ± calcite. Like the detrital grains, most of the additional magnetic phases, including hematite pigment and hematite coatings, contain rutile alteration. In site 5, more than 80% of diagenetic opaque grains are rutile/leucoxene.

The above observations suggest that hematitization took place before and during all stages of silicification and during later calcification. Also significant in site 5 is the lack of hematite in late fractures, thus relegating the hematitization to pre-Laramide times.

In summary, there is evidence for detrital hematite and magnetite/hematite grains, diagenetic martite, diagenetic hematite overgrowths, pre- and postdepositional hematite coatings on quartz, and at least three phases of hematite pigment that may be variously related to the postdepositional hematite coatings on quartz.

Magnetic components of Thundercloud Formation

The unresolved magnetic signature of this mixture of magnetic phases is depicted in Fig. 3b as a concentration of high-inclination directions to the northwest with a significant spur to the southwest. This distribution of NRM directions was separated into three different magnetic components (TH_A, TH_B, TH_C) by the various demagnetization treatments outlined (Table 2). TH_A is a shallowly inclined component with east- or west-orientation. TH_B is moderately inclined to the west, and TH_C is steeply inclined.

The ranges over which these components were removed or demagnetized by the various treatments and the interpreted host minerals are given in Fig. 6. Examples of the components revealed under the various treatments are shown in Fig. 7. Components from some specimens were not resolved because of the disintegration of specimens under acid leaching (sites 5, 7, 8, 16) or the acquisition of large induced magnetizations between 545° and 600°C under thermal treatment (sites 14–17, 21, 22).

TH_A is clearly a primary magnetization in site 13, where it is associated with magnetite and reversals, and probably at site 23, where there is also, by inference, magnetite, and a definite reversal stratigraphy. Considering the size of the sample collection, TH_A is not frequently recovered, with as few as one or two determinations at recorded sites (Table 2). Directions are somewhat scattered (Table 2), perhaps due to local rotations (see below); but a tilt test (McElhinny, 1964) on six site directions confirms that TH_A is pre-folding (pre-Laramide, see discussion in Jefferson and Parrish, 1989). Dispersion of site directions after ($k=10.12$) and before ($k=1.90$) correction for tilt yields a ratio (5.33) that, when compared to the appropriate value from the F-tables (3.13), indicates a positive test at the 99% confidence level.

The major component, TH_B, resides principally in diagenetic hematite that both coats quartz grains and is disseminated in the matrix. TH_B possibly also resides in minor phases such as various forms of acicular hematite, and as martite after detrital or diagenetic magnetite, as observed at site 5. Diagenetic quartz partly postdates the major diagenetic hematite phase. A fold test on six site directions of TH_B (Table 2), as with TH_A, also confirms a pre-folding magnetization at the 99% confidence level (variance ratio of 17.69 exceeds tabled value of 3.13).

The rarely recovered third component TH_C appears to predate some of the diagenetic quartz. It likely resides in a second phase of diagenetic disseminated hematite and of hematite that occurs in fractures. This second phase of diagenetic disseminated hematite

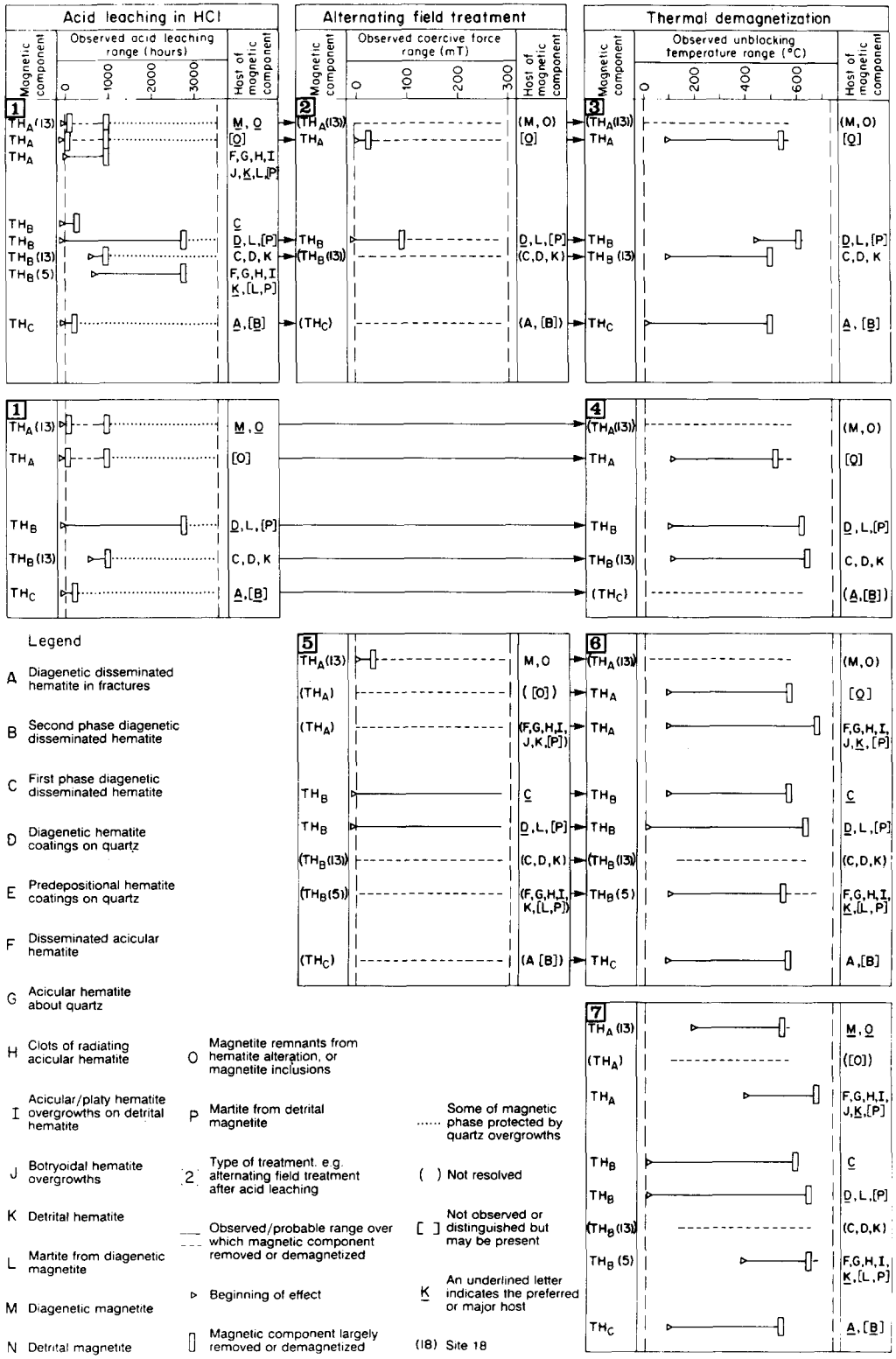


Fig. 6. Documentation of the characteristics of magnetic components from the Thundercloud Formation and the assignment of components to magnetic phases present. Magnetic components explained in the Methods section.

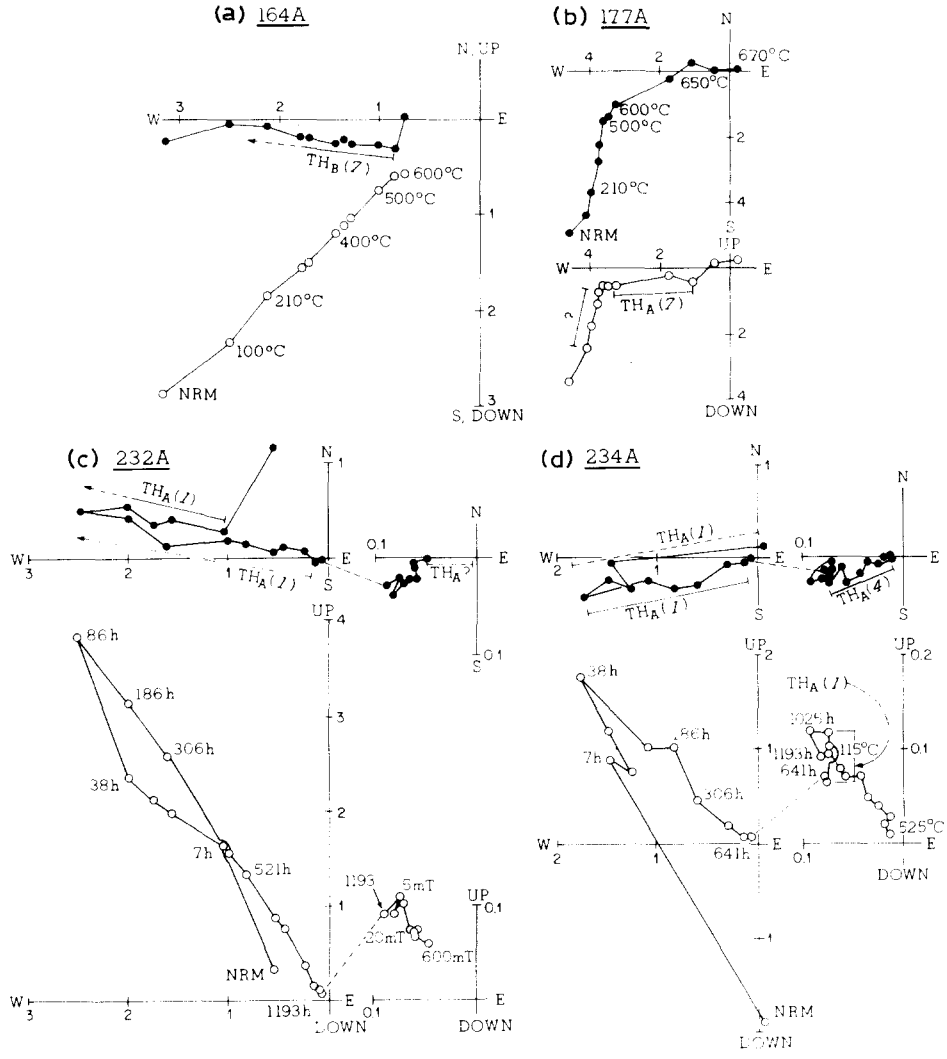


Fig. 7. Vector diagrams of the demagnetization of several Thundercloud specimens using a variety of treatments, singly and in combination. All vectors are plotted relative to the present horizontal. See Fig. 5 for further explanations.

cannot be visually distinguished from the first phase.

Petrography of Redstone River Formation

Sites from the Redstone River Formation are placed in stratigraphic context in Fig.2. They are located in thin proximal (RR2) and thick distal (RR3, RR4) alluvial fan facies that overlie sulphate evaporites (RR1). RR2 consists of planar bedded and trough cross-bedded conglomerates to sandstones that are composed mainly of carbonate grains with about 25% subangular quartz, feldspar, and lesser amounts of other lithic grains, such as chert. RR4 consists of fining-upward rhythmites of fine, micro-trough-crossbedded sandstone to laminated mudstone. The rhythmites have significant siliciclastic and volcaniclastic components which suggest different provenance from that of the proximal carbonate-dominated RR2 (Lustwerk, 1990). RR3 is similar to RR4 and contains numerous gypsum and anhydrite interbeds and laminae.

The complex magnetic mineralogy in RR2, 3 and 4 is similar to that of the Thundercloud Formation.

Rounded to angular altered detrital hematite grains, many with thin botryoidal hematite overgrowths, are concentrated in heavy mineral laminae. Some of these large hematite grains may be martite that originated from detrital magnetite incorporated in the sediment. Much hematite, particularly that in patchy irregular clots, clearly was derived by alteration of mafic detrital grains.

Hematite/goethite overgrowths are not as prominent as in the Thundercloud Formation. Some hematite rims about quartz appear to be continuous, others are discontinuous, as if abraded; these observations suggest that some rims are predepositional. Most hematite rims are clearly diagenetic, with the same crystal morphologies as matrix hematite. At sections 79-3 and 79-4 some silica overgrowths cover hematite rims.

Pigmental disseminated hematite varies in abundance. Some is included in the quartz overgrowths, other hematite is outside and may postdate the quartz. Pigmental hematite includes very fine-grained euhedra of hematite (martite) after magnetite and disseminated matrix hematite that is too fine-grained to resolve optically.

Rutile is common; it is intergrown with hematite, rims quartz and opaque grains, and is interspersed with hematite pigment.

Magnetic components of Redstone River Formation

The NRM distribution of Redstone River Formation (Fig. 3c) is similar to that of the Thundercloud Formation but lacks its prominent southwest spur (Fig. 3b). This distribution comprises at least two components. One component (RR_B) was identified by most of the treatment types (Table 3). Its direction (Table 2), similar to TH_B from the Thundercloud Formation, is likewise probably prefolding. The other major component, RR_F, has a similar direction to the west, but with a slightly shallower inclination. It is prefolding at the 99% level of the fold test (variance ratio of 4.48 compared to tabled value of 3.13; 6 sites). Some distinct outlier directions (RR_D, RR_E; Table 3) were also revealed. Components, treatment ranges, and probable magnetic hosts are set out in Fig. 8. Vector diagrams of components are shown in Fig. 9. Resolution of components was hampered by the acquisition of induced magnetizations beyond 625°C under thermal treatment (sites 1, 2, 20) and by the disintegration of specimens under acid leaching (sites 4, 19).

RR_F was revealed by acid leaching (Fig. 9e), thermal demagnetization (Fig. 9d), and in some specimens by AF treatment. It is interpreted to reside mostly in detrital hematite or martite (Fig. 8). RR_B, revealed prior to RR_F by all treatments, resides in diagenetic disseminated hematite. Components RR_D and RR_E probably reside in hematite pigment and in hematite coatings on quartz grains, respectively. These components may represent a rotated RR_B component: this possibility is discussed below.

Petrography of Coppercap Formation, basal member CP1

Only one site (25) was obtained from this member at section 79-3, but it comprises three rock types: red and green siltstones, and partly dolomitized laminated microbial limestone. The limestone contains disseminated chalcopyrite, concentrated along microbial laminations. Several magnetic phases were documented in these rocks.

Pigmental hematite is abundant in the red siltstone but very rare in the green siltstone.

Detrital hematite, up to 40 μm in size, was positively identified in the red siltstone. No detrital hematite is preserved in the limestone, but pyrite grains are partly or wholly altered to goethite where surface weathering has affected

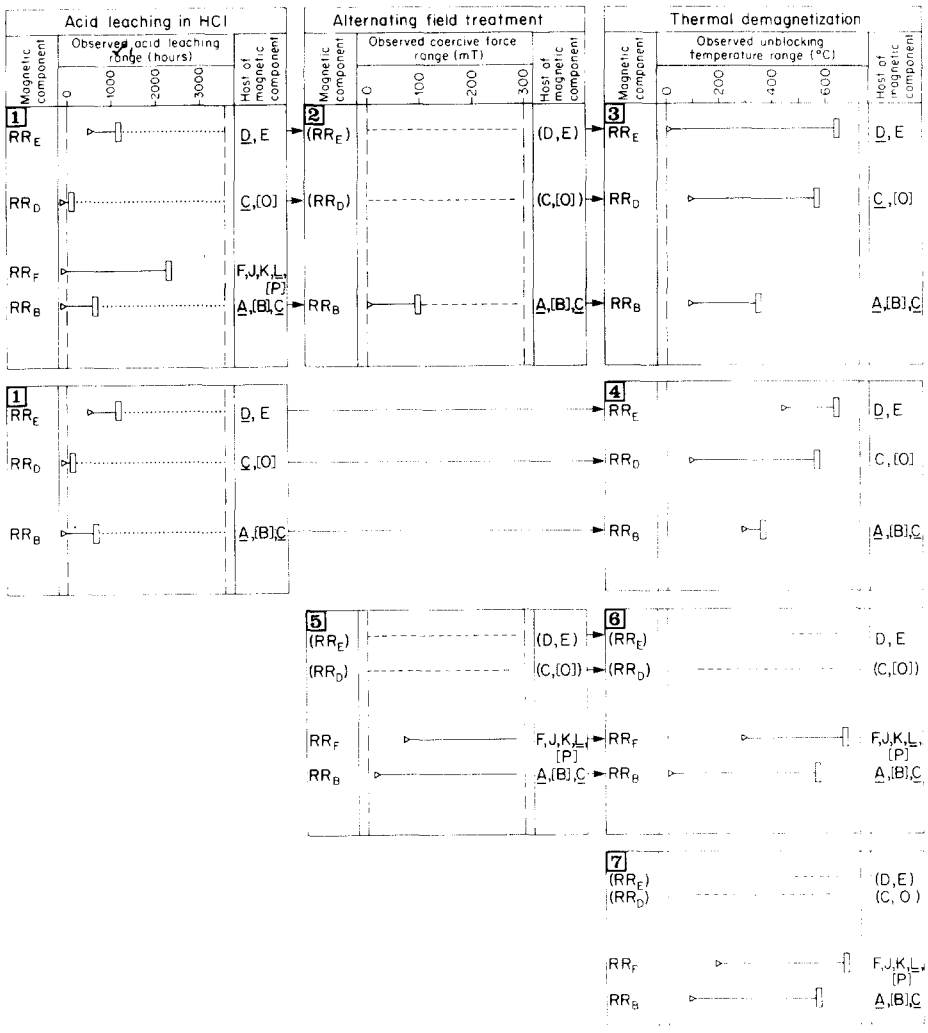


Fig. 8. Documentation of the characteristics of magnetic components from the Redstone River Formation and the assignment of components to magnetic phases present. Legend in Fig. 6. Magnetic components explained in the Methods section.

the rock. In the green siltstone and microbial limestone, most detrital hematite is altered to or mantled by rutile.

Magnetic components of basal Coppercap Formation

NRM directions are scattered about the PEF (Fig. 3d) with associated intensities that range from 5 to 10 mA/m for the red siltstone, 0.5 mA/m for the green siltstone, and 0.05 to 0.4 mA/m for the limestone. To resolve the NRM, the red siltstone was treated by CH and TH methods and the limestone and green siltstone by AF and TH. Three components (CP_A, CP_B, CP_C) were resolved (Table 4).

In the red siltstone, TH treatment revealed component CP_B with a direction similar to RR_B and TH_B (Fig. 10a). CH treatment failed to resolve the NRM: although it was discontinued after 300 hours due to the presence of an induced magnetization, data suggest the removal of CP_B.

In the limestone and green siltstone, AF and TH treatments (Fig. 10b, 10c) revealed a

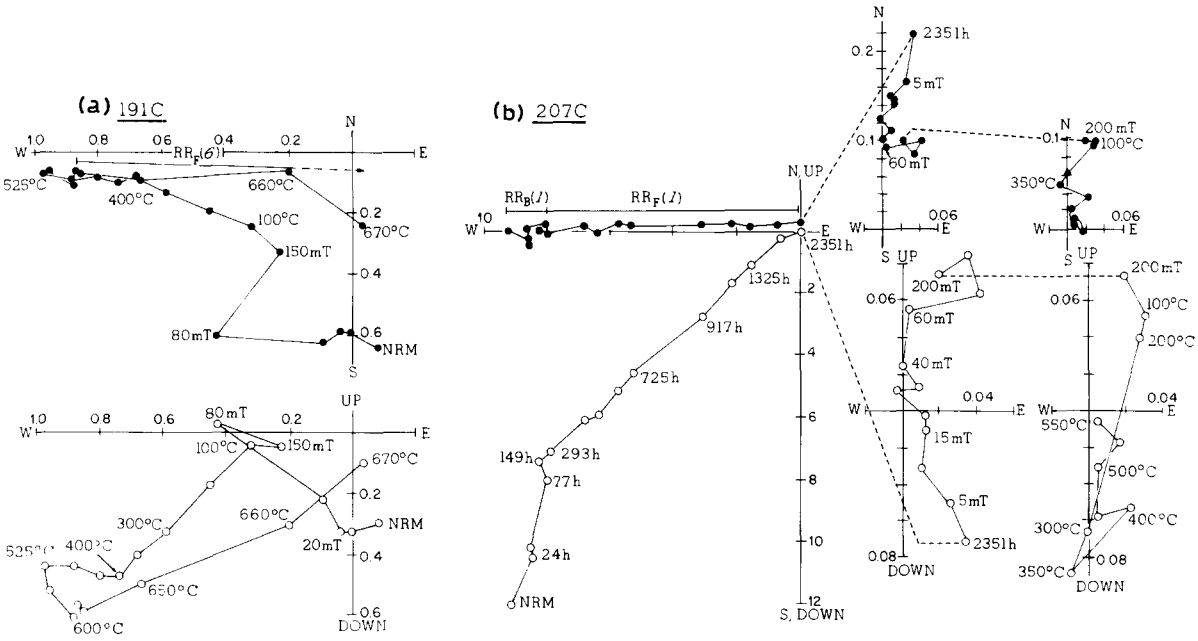


Fig. 9. Vector diagrams of the demagnetization of several specimens from the Redstone Formation using a variety of treatments. All vectors are plotted relative to the present horizontal. See Fig. 5 for further explanations.

TABLE 4
Mean component directions from site 25 of the Transition Zone, Coppercap Formation

Comp.	Sa.	Sp.	Det.	Treatm.	$D°,I°$ (w.r.t. hor.)	$D°,I°$ (w.r.t. bdg.)	R	k	$\alpha_{95}°$	Pol.
CP_A	*4*	11	11	5, 7	035, +74	295, +46	3.97	90	10	N
CP_B	1	*3*	3	7	100, +87	275, +42	2.99	283	7	N
CP_C	1	*3*	3	5	087, +47	313, +81	2.94	32	22	N

See Table 1 for explanation of symbols. Italics indicate unit weight. Means plotted in Fig. 10.

component CP_A with a closely similar direction to CP_B (Fig. 10a).

In the early stage of AF treatment the limestone also exhibited a component CP_C, having a high inclination when corrected for tilt.

Toward the end of treatment in all rock types, the direction of the remanent magnetization suggests the presence of an unresolved westerly directed component, which has a lower inclination than CP_A (Fig. 10b). However, several factors probably prevented its resolution: (1) the onset of induced effects during heating; (2) the introduction of rotational remanent magnetization (RRM) in AF; and (3) the presence of noise in the measurement of weakly magnetized samples. Induced magnetizations became significant above 640°C in the red siltstones, above 550°C in the green siltstones (Fig. 10b), and above 300 to 400°C in the limestones. RRM, which is particularly acquired by the algal limestones, increases in magnitude with progressive AF treatment (Fig. 10c).

Magnetic characteristics of CP_A (remanent coercive forces < 100 mT, unblocking temperatures < 550°C) suggest that it resides in magnetite in the green siltstone, though magnetite is not observed petrographically. Hematite pigment with unblocking temperatures less than 550°C appears to contain the CP_B com-

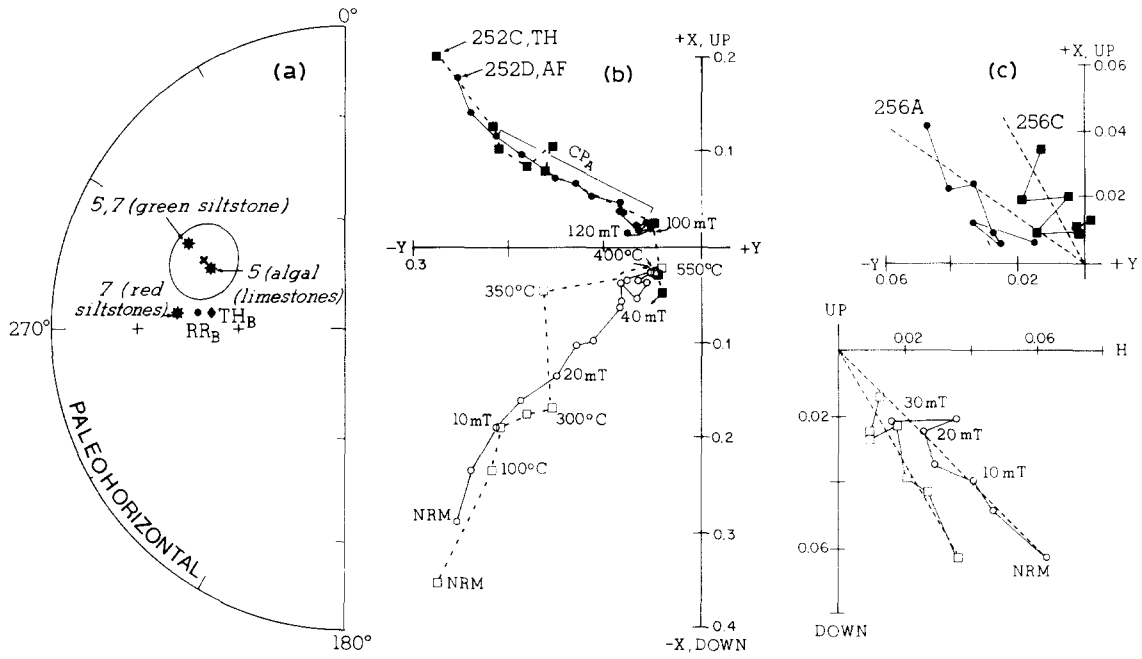


Fig. 10. Magnetic components from the Transition Zone, section 79-3. (a) Removed magnetizations from site 25 compared to other B components from the Coates Lake Group. Italic numerals refer to treatments (Table 4). Vector diagrams showing AF and thermal treatment of specimens of green siltstone (b) and algal limestone (c). All vectors are plotted relative to the paleohorizontal. See Fig. 5 for further explanations of vector diagrams.

ponent in the red siltstone; but demagnetization data suggest that detrital hematite could also be a carrier, in common with TH_B and RR_B. It is possible that the unresolved low-inclination component to the west also resides in hematite because the unblocking temperatures are greater than 550°C. The host of CP_C is not known.

Magnetic components of site 24, Little Dal lavas

A basic igneous unit, lying just north of the Redstone River (126.6°W, 62.9°N), was assigned to the Little Dal lavas by Jefferson and Parrish (1989). AF treatment of samples from this unit revealed a mainly mixed magnetization comprising at least three components, only one of which could be reliably resolved. Endpoint directions from two specimens and specimen demagnetization data from four specimens using the "remagnetization circle" method (Halls, 1976) yielded a direction of D, $I=023°$, $+53°$ ($k=64$, $\alpha_{95}°$ max$=13°$; Bailey and Halls, 1984) with respect to the present horizontal and 068°, +78° with respect to the paleohorizontal.

Discussion

Paleopoles

Paleopoles of this study (Table 5) are somewhat dispersed from previously reported poles (Table 6), as shown in Fig. 11. Primary poles (LR_A, TH_A, CP_A?) and early-diagenetic hematite pigment poles (TH_B, RR_F, RR_B, RR_E) tend to lie on or to the west of a previously proposed apparent polar wander (APW) path (Fig. 11) (Park and Aitken, 1986a, b). Late-diagenetic hematite pigment poles (LR_{C1}, LR_{C2}, TH_C) lie to the north (Fig. 11c, path 3) or south, as antipoles (Fig. 11a, path 2) in high latitudes. The high-latitude poles probably relate to a later APW path, either in the later Precambrian or in the Cretaceous.

TABLE 5
Summary of paleomagnetic poles

Component	Sites	N	Pole (°W, °N) (in situ)	Pole (°W, °N) (paleohorizontal)	R	K	A_{95}°
Rusty Shale formation							
1. LR_A	9, 11, 12	3	–	227, −08	2.96	52	17
2. LR_{C1}	9–12	4	065, +68	–	3.72	11	30
			–	226, +71	3.79	14	25
3. LR_{C2}	10–12	3	037, +27	–	2.93	28	24
			–	032, +49	2.86	15	34
4. LR_{C3}	18	1	130, +72	168, +23	–	–	–
Little Dal basalt, site 24							
5. LL_C	24	1	345, +58	076, +63	–	–	–
Thundercloud Formation							
6. TH_A	7, 8, 13, 16, 21, 23	6	–	210, +01	5.56	11	21 [a]
7. TH_B	5–8, 16, 23	6	–	202, +27	5.91	53	9
8. TH_B	13–15	3	–	205, +11	2.93	29	23
9. TH_C	21, 22	2	189, +14	–	1.99	–	–
			–	015, +72	1.98	–	–
Redstone River Formation							
10. RR_F	1–4, 19, 20	6	–	209, +22	5.92	59	9
11. RR_B	1–4	4	–	206, +32	3.97	92	10
12. RR_D	1–3	3	–	233, +39	2.98	88	13
13. RR_E	1–3	3	–	258, +41	2.99	162	10
Coppercap Formation, Transition Zone							
14. CP_A	25	1	–	225, +35	–	–	–

The mean poles given are calculated from the site poles, where N is the number of sites, R is the vector resultant, K is the precision, and A_{95}° is the half angle of the cone of 95% confidence about the mean. All site poles except entry 5 are derived from data given in Tables 1–4. Data under entry 8 may reflect unknown tilting (see Table 2, text).
[a] Large error in entry 6 may be due to differential rotation at localities.

The western poles can be interpreted in two ways. First, Park and Aitken's (1986a, b) APW path is defined incorrectly, and the northward-trending APW segment should incorporate these poles (Fig. 11a, path 1). Alternatively, the discrepant poles record rotational tectonic effects that can be corrected for (e.g. Figs. 11b, 11d). Before the interpretation of the APW path is presented, together with its implications, we first discuss the possibility of tectonic corrections and then consider the possibility of Late Precambrian high-latitude poles.

Tectonic effects on poles (Figs. 11, 12)

Morris and Park (1981) originally suggested that the APW path might extend northwesterly to include pole LL_{A1} from the Little Dal lavas, as shown in alternative path 1 of Fig. 11a. The resolution of poles RR_E and RR_D in this study lend weight to the argument that the APW path should be extended from S_A, the primary diabase pole (Table 6), through LL_{A1}, RR_E, and RR_D to poles representing the hematite pigmentation event P2. Relative ages of all these poles appear to be in temporal order along the path. RR_E and RR_D are considered to predate RR_B and would postdate LL_{A1}, if LL_{A1} is a primary pole.

Park and Aitken (1986a) preferred to reject LL_{A1} as a primary pole, assigning it instead to

TABLE 6
Paleopoles from previous studies used in this paper

Symbol	Description	Sites	Pole (°W,°N)	$\delta p°, \delta m°$	$A_{95}°$	Reference plane	Reference
FD_N	Franklin diabases, normal	27	196,09	–	5	Paleohor.	[1–4]
FD_R	Franklin diabases, reverse	51	198,02	–	3	Paleohor.	[1]
K_A	Katherine Group	5(13) [a]	210,09	(4, 8)	–	Paleohor.	[8]
K_B	Katherine Group	4	196,17	4, 6	5	Paleohor.	[8]
K_C	Katherine Group	9	122,77	11, 12	11	Pres.hor.	[8]
LB_A	Basinal sequence, Little Dal Group	14	219,–16	3, 5	3	Paleohor.	[9]
LB_B	Basinal sequence, Little Dal Group	13	222,–03	5, 11	8	Paleohor.	[9]
LB_C	Basinal sequence, Little Dal Group	12	190,60	8, 9	9	Paleohor.	[9]
LL_{A1}	Basalts, Little Dal Group	4	246,24	4, 7	5	Paleohor.	[10]
LL_{A2}	Basalt, Little Dal Group	1	223,11	–	–	Paleohor.	[11]
LL_B	Basalts, Little Dal Group	6	201,24	7, 11	12	Paleohor.	[10,11]
LL_C	Basalts, Little Dal Group	10	287,87	–	9	Pres.hor.	[10–12]
LM_A	Mudcracked formation, Little Dal Group	3(7) [b]	217,-09	(6, 12)	–	Paleohor.	[13]
LM_B	Mudcracked formation, Little Dal Group	1	196,24	–	–	Paleohor.	[13]
NB	Natkusiak basalts	14	201,06	4, 8	–	Paleohor.	[14]
RP	Reynolds Point Formation	10	213,–06	4, 7	–	Paleohor.	[14]
R_X	Rapitan Group	19	166,–07	–	~5	Paleohor.	[15]
$R_Y(N)$ [c]	Rapitan Group	(81) [a]	163,53	(7, 7)	–	Paleohor. [d]	[15]
$R_Y(R)$ [c]	Rapitan Group	(53) [a]	059,75	(14, 14)	–	Pres.hor. [d]	[15]
R_Z	Rapitan Group	17	211,05	–	~5	Paleohor.	[15]
S_A	Diabase sills and dykes	17	222,02	–	5	Paleohor.	[16,17]
S_C	Diabase sills and dykes	5	140,66	–	16	Pres.hor.	[17]
T_A	Tsezotene Formation	14(23) [a]	214,12	(5, 9)	–	Paleohor.	[18]
T_B	Tsezotene Formation	18	198,23	3, 5	5	Paleohor.	[18]
T_C	Tsezotene Formation	29	141,69	6, 6	–	Pres.hor.	[18]

$\delta p°, \delta m°$ are the semi-axes of the ellipse of confidence at probability level 0.05 about the pole calculated from the average direction. $A_{95}°$ is an alternate error explained in Table 5. Reference plane for calculation of paleomagnetic directions is the present horizontal (pres. hor.) or the past horizontal (paleohor.). References: [1] Fahrig et al. (1971); [2] Fahrig and Schwarz (1973); [3] Park (1974); [4] Christie and Fahrig (1983); [5] Robertson and Baragar (1972); [6] Palmer and Hayatsu (1975); [7] Park (1981c); [8] Park and Aitken (1986a); [9] Park (1981a); [10] Morris and Aitken (1982); [11] Park, unpublished data; [12] this study; [13] Park (1974); [14] Palmer et al. (1983); [15] Morris (1977); [16] Park (1981b); [17] Park et al. (1989); [18] Park and Aitken (1986b).

[a] Pole calculated from specimen directions (in parentheses).

[b] Pole calculated from sample directions (in parentheses).

[c] Normal (N) and reverse (R) R_Y components have been estimated from specimen directions plotted in fig. 2 of Morris (1977). R_Y is found in 35 sites, but site statistics of normal and reverse components are not given.

[d] Suggested reference plane (see text).

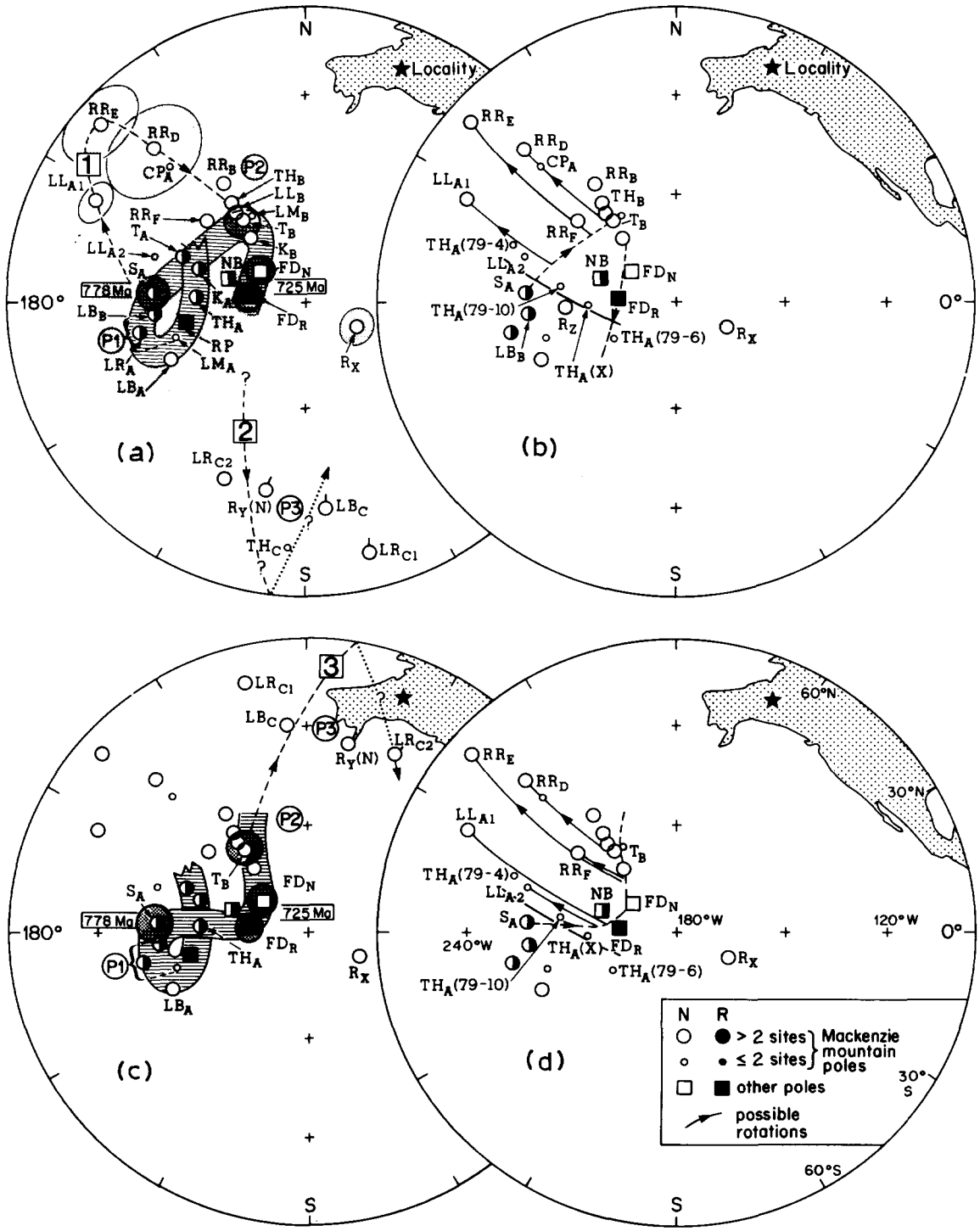


Fig. 11. Apparent polar wander (APW) paths derived mainly from units of the Mackenzie Mountains. See text for a discussion of the major alternative APW paths given in (a) (after Park and Aitken, 1986a) and (c), and variations (dashed lines) indicated by numbers 1, 2, and 3. Possible local rotations about the mean locality (star) are shown in (b) and (d) for each major alternative APW path. Pole errors (A_{95}°) are shaded. Errors of reference poles have darker shading. Poles (from other studies) plotted on the lower hemisphere are indicated by a projected line on the symbol; the possible corresponding path is a fine dotted line. Locality poles for TH_A are plotted in (b) and (d). They are calculated using specimen unit weight. P1, P2, and P3 refer to hematite pigmentation events. Poles from other studies plotted here and in Fig. 13 are listed in Table 6. See Table 5 for a summary of the pole data of this study.

some later period and therefore to a different APW segment. Arguments to support their rejection of path 1 (Fig. 11a) are: (1) LL_{A1} should plot close to S_A if the lavas and diabase represent the same igneous episode; (2) the primary Thundercloud pole TH_A does not lie on extension 1, whereas from stratigraphic considerations it should lie between LL_{A1} and poles of the Redstone River Formation; and (3) geological evidence of syn-sedimentary and syn-diagenetic block faulting favours the explanation that discrepancies are due to local tectonic rotations.

Let us consider the paleomagnetic evidence for displacement of poles by tectonic activity; namely, by geologically undetected tilting or rotation. For reference, we have used poles FD_N and FD_R from the craton, and S_A and T_B that reside in widely distributed units of the Outer Fold Belt (Table 6). S_A and T_B reside in strata that contain no geological evidence of syn-depositional faulting. Furthermore, comparisons between results from different thrust sheets have not revealed rotations (Park and Aitken, 1986b; Park et al., 1989).

Geological evidence for Late Precambrian folding is controversial, and limited to the Nite Creek area (Helmstaedt et al., 1979; Aitken et al., 1981) and to the area just north of section 79-10 (Jefferson and Parrish, 1989); however, no paleomagnetic samples were taken from these areas. Paleomagnetic evidence for folding, apart from that associated with the Laramide Orogeny, is not readily detectable in any of the poles reported here. Any pre-Laramide folding would affect the calculation of all poles predating that folding, except primary poles of units later folded about the same axis.

Conglomerates and facies changes clearly indicate that Late Proterozoic faulting affected all strata within and below present exposures of the Coates Lake Group and Sayunei Formation, i.e. the entire area of the Plateau Thrust system. Block tilting could have resulted from listric extensional faulting, which is common in rift zones and which in some cases results in tilts on the order of 30° or more (e.g. Morretti and Colletta, 1988). Some tilting may be simulated by rotations about inclined axes and may be achieved in a variety of ways (Macdonald, 1980).

Despite the geological evidence of faulting throughout the Coates Lake Group, in most places the amount of tilting appears to have been minimal. Complexities associated with Late Proterozoic tilting are only evident at geological section 79-10 (sites 13-15), which is surrounded by fault blocks displaying variable attitudes. Because of this complexity, only the probable primary magnetic component from site 13 is used for pole determinations. As far as can be determined, the tilting at the remaining geological sections sampled for this study is due to Laramide folding about horizontal axes.

If tilting does not cause the aberrant poles, tectonic rotation must then be considered. Clockwise rotations suggested by these poles are compatible with: (1) the extensive Proterozoic fault systems that have been mapped and inferred (Eisbacher, 1981; Jefferson, 1983; Jefferson and Ruelle, 1986; Jefferson and Parrish, 1989); (2) the hypothesis of dextral strike-slip fault movement in the Late Precambrian (e.g. Eisbacher, 1977; Bell, 1982); and (3) the proximity of the region to a possible Proterozoic ridge-transform intersection (Yeo, 1984; Allerton, 1989).

Some minor dextral and sinistral rotations of Laramide age may be associated with thrusting along curved arcs. For example, geological sections within the Plateau Thrust panel (X, 79-8, 79-9, 79-4) may be rotated slightly counterclockwise or clockwise relative to the pinned points at the terminations of the fault (J.D. Aitken, pers. commun., 1990). These rotations are considered too minor to account for the pole dispersions reported here.

Rotations are especially difficult to document by geological methods. However, paleomagnetism can strongly suggest rotations from a distribution of component directions about

small circles. Several of the poles reported here are distributed in this way, compatible with significant clockwise rotations of blocks within the Plateau Thrust zone, especially about local axes.

The most convincing paleomagnetic evidence for rotation is found at Redstone River sites within the unbroken section 79-3, where a minimum 52° clockwise movement is inferred to have taken place during magnetization (Fig. 12a). Relatively old magnetic components (RR_D, RR_E) in this section reside in diagenetically altered framework phases. Relatively young magnetic components (RR_B, still pre-Rapitan Group) have been resolved in disseminated hematite pigment. A comparison of TH_A directions from the various localities is also suggestive of rotation (Fig. 12b): they lie roughly along a possible small circle arc of about 35°.

Similar rotation effects are also suggested from a comparison of P2 poles (Fig. 11a);

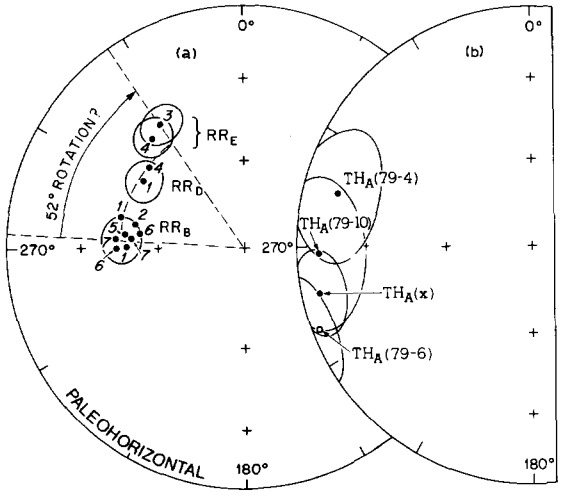


Fig. 12. Evidence for possible local tectonic rotation at Redstone River Formation, section 79-3 (a) and at Thundercloud Formation sections (b). Numbers in italics represent treatments as shown in Fig. 9. Errors (α_{95}) are indicated by ellipses. RR_D and RR_E errors shown in Table 3 result from combining data from several different treatments. Means are downward directed (solid circles) or upward directed (open circles).

poles are distributed in a line subparallel to a small circle centred at the mean locality within Mackenzie Mountains. Poles from the Plateau Thrust Sheet (RR_B, TH_B, LL_B) lie northwest of poles from the Outer Fold Belt (T_B, LM_B, K_B), and this arrangement is compatible with the latter being fixed references and the former being rotated clockwise.

All rotations documented and inferred above involve units located within the Plateau Thrust system. Given that local rotations took place within the rift sequence, tentative corrections can be made to anomalous poles from previous studies to check on the reasonableness of this explanation. Pole LL_{A1} from the Little Dal lavas (Morris and Aitken, 1982) has always appeared to be anomalous. If it is rotated counterclockwise about the locality position, it passes very close to a basalt pole (LL_{A2}) (Park, unpublished) that lies near poles (S_A, LB_B) from the Outer Fold Belt (Fig. 11b). If LL_{A1}, LL_{A2}, and locality TH_A poles are rotated toward S_A, they cluster in a relatively small group, which would be expected if the basalts were fed by the diabases. This would require clockwise rotation of all but one of the TH_A poles; but this is not consistent with the counterclockwise rotations inferred above.

Alternatively, LL_{A1} and LL_{A2} may be rotated to the vicinity of FD_N, FD_R and NB (Fig. 11d), which represent the 723 ± 3 Ma (Heaman et al., 1990) Franklin igneous episode (Fahrig and Schwarz, 1973). This interpretation would require that the Little Dal basalts be some 50 Ma younger than the diabase intrusions.

In summary, consideration of possible Late Proterozoic tectonic effects to the poles has suggested that differences amongst poles are more explainable by rotations than by folding or tilting. In general, for A (primary) poles of this study, the effects of both Cretaceous and possible Proterozoic compressional folding would be simply compensated by correction for bedding. For B (diagenetic hematite pigment) poles, the effects of folding would be unknown

if folding occurred prior to the acquisition of the B magnetization. Because of excellent agreement amongst P2 poles (Fig. 11a) when they are corrected for possible rotational effects, it is probable that there is little geologically undetected Late Proterozoic folding at any of the sampling sites.

Sample sections have been chosen to avoid structurally complex areas. One section, 79-10, however, which was later found to have structural complications (the Proterozoic folds described by Jefferson and Parrish, 1989), yields "B" poles with lower inclination than the P2 poles (Table 2). These secondary poles tend to support the hypothesis of Proterozoic tilting and (or) folding given by Jefferson and Parrish (1989), but are not used in any of the figures because the combination of Proterozoic and Cretaceous structural deformation could not be resolved.

Significance of high-latitude poles

High-latitude poles from Precambrian units of the Mackenzie Arc (Fig. 13) determined from this study (LR_{C1}, LR_{C2}, LR_{C3}, TH_C) and others (LB_C, LL_C, K_C, T_C, S_C, R_Y; see Table 6) are not readily incorporated into an APW path. Morris and Park (1981) interpreted a path that passed over the North Pole and included results from outside the Mackenzie Arc; but there is little age control, and the starting point for this path is the much older Grenville Loop (greater than 820 Ma or 880 Ma, Park and Aitken, 1986a). One high-latitude pole from the Mackenzie Arc, T_C, definitely post-dates Laramide folding, and relates either to the Cretaceous–Paleocene Laramide Orogeny (about 65 Ma) or to the Recent Epoch. The major question is: Are all these high-latitude poles Laramide or younger in age or are some older, possibly Late Precambrian?

All high-latitude poles are plotted in Fig. 13. Poles from the Outer Fold Belt tend to form a tighter group (Fig. 13a) than those of the Plateau Thrust Sheet (Fig. 13b). The former are separated into postfolding and prefolding directions, that is, according to whether precision is greater before or after correction for tilt. Thus, LB_C is prefolding (at 99% significance level; Park, 1981a) and T_C (Park and Aitken, 1986b) and S_C (Park et al., 1989) are postfolding (at 95 and 90% levels, respectively). K_C is syn-folding, showing the greatest precision when beds are unfolded by 40%; though the fold-test at this stage is not significant at the 95% level.

There is no tilt or fold test for most high-latitude poles of the Plateau Thrust Sheet. LL_C is clearly postfolding at the 95% level of the tilt test, because the calculated variance ratio of 1.95 from ten site poles (8 sites, Morris and Aitken, 1982; 1 site, Park, unpublished data; 1 site, this study) exceeds the value of 1.78 from *F*-tables. LR_{C3} from the present study also appears to be postfolding: its pole, uncorrected for bedding, lies amongst postfolding poles of the Outer Fold Belt (Fig. 13c), and it appears to be associated with hematite filling late fractures attributed to the period of the Laramide Orogeny. But other poles from the present study could be prefolding in age. A tilt test performed on poles LR_{C1}, LR_{C2}, and TH_C reveals that the calculated variance ratio of 4.54 exceeds the tabled value of 4.28, thus suggesting that these poles are prefolding at the 95% level. The prefolding case is further supported because two of the poles (TH_C, LR_{C2}) lie at low latitudes when uncorrected for bedding (Table 5); this postfolding case is clearly not possible, unless the poles record transitional fields during periods of reversal or attempted reversal. One other pole from the Plateau Thrust Sheet, R_Y, is not clearly either pre- or post-folding in age. An examination of R_Y directional data (Morris, 1977, fig. 2) reveals that normal ($R_Y(N)$) and reverse ($R_Y(R)$) polarities could have distinctly different directions. $R_Y(N)$ predates $R_Y(R)$, and they are tentatively plotted as pre- and post-folding directions, respectively, in Fig. 13c.

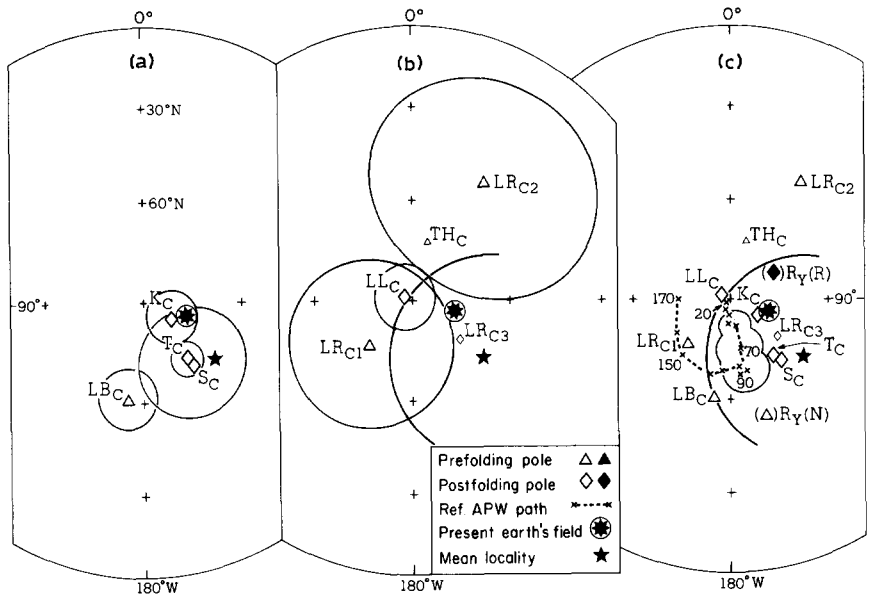


Fig. 13. High-latitude paleopoles from the Mackenzie Mountains plotted on a polar projection: (a) Outer Fold Belt, (b) Plateau Thrust Sheet, (c) combined. Poles in (c) may be compared to an APW reference path for the period Middle Jurassic to present (Irving and Irving, 1982) or to a locus representing possible local rotation in the Late Precambrian about the locality mean. Numbers represent millions of years. Errors of reference poles representing the period Late Cretaceous to Paleocene (the period of the Laramide Orogeny in the region) are shown in (c). Small symbols represent poles calculated from one or two sites. Open and closed symbols refer to normal and reverse poles. Poles from this study are summarized in Table 5 and poles from previous studies are listed in Table 6.

Even though some poles are prefolding, they are not necessarily Late Precambrian; associated errors overlap Cretaceous poles, suggesting that they could be Cretaceous in age. Nonetheless, poles ascribed to prefolding mostly lie in different sectors of Fig. 13c than poles associated with postfolding, and they are more dispersed. Dispersal may be caused by local rotations due to faulting either in Cretaceous or Late Precambrian times. A possible locus of rotation is shown in Fig. 13. Local rotations have been proved to be present for Late Proterozoic units of the Plateau Thrust Sheet, suggesting that the dispersal of high-latitude prefolding poles also relates to local rotations occurring in the Late Precambrian. These poles could be constrained to the Late Precambrian because quartz cement, which postdates P3 components of this study, is possibly derived from hydrothermal systems operative during Rapitan deposition (Yeo, 1981) or later.

Apparent polar wander path

The APW path is anchored by poles S_A, T_B, FD_N, and FD_R (Fig. 11). S_A is dated at $777.7^{+2.5}_{-1.8}$ Ma (U–Pb, zircon, Jefferson and Parrish, 1989) and 766–769 ± 27 Ma (Rb–Sr, Armstrong et al., 1982); FD_N and FD_R are dated at 723 ± 3 Ma from preliminary U–Pb studies of baddeleyite (Heaman et al., 1990) in sills of the Franklin diabases. Two major alternative APW paths may be constructed through the key poles. The first alternative (A) (Fig. 11a) (Park and Aitken, 1986a, b) is ordered $S_A \rightarrow T_B \rightarrow$ FD; the second (B) (Fig. 11c) is ordered $S_A \rightarrow$ FD $\rightarrow T_B$. The main advantage of (A) is that it has fewer bends. The main advantage of (B) is that it appears to better correlate the physical separation of poles S_A and FD with their temporal separation (about 50 Ma) and bridges them by TH_A, which likely postdates S_A. Also, if we consider

TH_A to have been rotated clockwise in common with other units from the Plateau Thrust Sheet and in accordance with geological structures, it fits better with alternative (B).

The route of the APW path beyond FD_R (Fig. 11a) or T_B (Fig. 11c) is more speculative. For (A) it could proceed south to pass through the P3 poles plotted as antipoles (path 2, Fig. 11a), if those poles are Late Precambrian. A second possibility is that it could proceed directly from T_B through the P3 poles, plotted in the north, to the Franklin antipoles (path not shown). However, this interpretation is less likely because of the extremely rapid apparent polar wander that it requires. A third possibility is that the path again proceeds from T_B to the northern P3 poles and then back to the Franklin poles (path not shown). Again, this requires rapid apparent polar wander. For (B) the path could proceed from T_B to the P3 poles, plotted in the north (path 3, Fig. 11c). Outlying poles may be explained as due to tectonic rotations as discussed (Figs. 11b, 11d).

The APW path currently does not incorporate pole R_X, which likely represents primary Rapitan deposition. One could argue that R_X has been rotated similarly to many other poles derived from the Plateau Thrust Sheet; however, its anticlockwise sense of rotation would be opposite to that of the other poles and incompatible with the geological evidence, which suggests right-lateral faulting (Eisbacher, 1981). Right-lateral faulting should result in clockwise rotations of blocks which would behave as bearings between sliding terranes. A current paleomagnetic study of the Sayunei Formation may explain this pole.

Paleomagnetic data from the Coates Lake Group suggest that the hiatus between the Thundercloud Formation and the underlying Little Dal Group was short-lived, whether we accept that the Thundercloud (TH_A) and Little Dal lava (LL_{A1}) poles have been rotated from APW path (A) (Fig. 11b) or from (B) (Fig. 11d). This is compatible with the low-angle unconformity and similar paleoclimates noted by Jefferson and Parrish (1989) between the Little Dal and Coates Lake Groups. Instead, there may be a longer hiatus within the Coates Lake Group itself, as indicated by the angular distance between TH_A and RR_E when both are rotated back to the APW path (Figs. 11b, 11d). Abrupt contacts and local breccias suggest an un-mapped discontinuity within the Redstone River Formation between member RR1 (evaporites) and RR2 (overlying conglomerate) (Jefferson and Ruelle, 1986).

Hematitization and mineralization of the Coates Lake Group

Petrographic studies of cupriferous carbonates of member CP1 (e.g. Chartrand et al., 1989; Lustwerk, 1990) have documented a paragenetic sequence that has not been directly linked to the paragenesis of red siltstones in the Redstone River Formation (ibid.; this paper). Chartrand et al. (1989) have shown that the copper mineralization in CP1 was an early diagenetic process. Lustwerk (1990) has shown by stable isotopes that the copper mineralization was completed by the time all of CP1 was deposited, and by trace elements that the red siltstones are markedly depleted in copper relative to other elements. These results have supported previous speculations (e.g. Jefferson and Ruelle, 1986) that copper was liberated from mafic detrital constituents by diagenetic hematitization and chloride complexing into formational brines, then moved upward through CP1.

Paleomagnetism provides a positive test and new perspectives on the above-hypothesized linkages between hematitization and copper mineralization. The most pervasive, paleomagnetically resolvable, hematite phases in the Coates Lake Group were apparently formed before deposition of the Rapitan Group, because most of the poles in Coates Lake Group sites predate, or are assignable to the P2 hematitization event, which is not recorded by the Rapitan Group. High-latitude P3 poles are

determined at only scattered sites below the Rapitan Group, and indicate a poorly resolvable hematitization that probably formed during/after deposition of the Rapitan Group. P2 hematitization recorded in the Coates Lake Group also could not have formed during Coppercap sedimentation above CP1, because the depositional environment was fetid and subaqueous. It is extremely unlikely that P2 is Cretaceous or Recent in age, because of the position of the P2 poles. Therefore the major P2 hematitization in Coates Lake Group was completed by the time the last red beds (member CP1) had been deposited.

In the following, we will first discuss the detailed paragenesis and magnetization of the red siltstones in RR3/4 and CP1 then that of the green and grey beds in CP1.

Paragenesis of RR3/4

Jefferson and Ruelle (1986) discussed field evidence for hematitization of basalts and basalt fragments both before and after incorporation in Coates Lake Group strata. Lustwerk (1990) and this study have determined that abundant diagenetic hematite pigment is present, as indicated by the microscopic background texture and colour of disseminated hematite, the hematite-altered mafic minerals, the inclusion of hematite in quartz overgrowths, the leaching experiments reported here, and studies of Recent reddening in sediments (e.g. Zeilinski et al., 1983). In particular, paleomagnetic components separated by acid leaching and distinguished by tectonic rotation definitely establish the red coloration as primarily due to the P2 diagenetic hematitization.

Paragenesis of CP1

Within member CP1, copper-bearing microbial carbonate and evaporite beds are bounded above and below by green, chloritic siltstones grading to red siltstones. Both red and green siltstones are compositionally and texturally very similar to the Redstone River siltstones. Based on preservation of red cores within concretions in CP1, Jefferson and Ruelle (1986, p. 158) concluded that the green chloritic siltstones were red before being reduced by "diagenetic chemical interaction between the fetid algal limestones and hematitic" siltstones. There is no evidence of subsequent reddening of the green beds, therefore the diagenetic hematite must have formed before the last reduction. The reduction of the siltstones also appears to post-date the event of uniform copper mineralization of continuous (>2000 m by >2000 by 1 m) limestone beds, because copper sulphides are redistributed irregularly (1–2 m lateral change) both above and below each copper-bearing limestone bed, especially along small fractures and minor faults (Jefferson and Ruelle, 1986).

Petrographic studies by Chartrand (Chartrand et al., 1989) have documented that, within the copper-bearing sabkha carbonates of the CP1 member, the preserved paragenetic sequence is:

detrital mineral laminae
microbially laminated primary carbonate minerals
diagenetic dolomite
sulphate + calcite + dolomite
framboidal + recrystallized pyrite
quartz + feldspar
copper-iron sulphides
euhedral pyrite + chlorite + calcite.

The reduction and chloritization of siltstones, and copper redistribution noted by Jefferson and Ruelle (1986), are interpreted to correlate with Chartrand et al.'s (1989) late diagenetic pyrite+chlorite+calcite phases. These could be very late, perhaps linked to low-grade regional metamorphism of the copper belt. This interpretation is the only way to reconcile the symmetrical local zoning, shown by the late phases, with the asymmetrical (upward-decreasing Cu/Fe ratio) zoning, shown by the overall 100-m CP1 member and by other similar copper deposits in the world (Kirkham, 1989). No evidence of the expected "rote-

faule" hematitization immediately beneath the zone of highest Cu/Fe has been recognized by any studies of the Redstone copper belt. Thus the late chloritic phase is thought to have overprinted and obscured some details of the primary zoning.

Sequence of depositional and mineralization events

Hematitization and sulphide mineralization events within the Coates Lake Group at section 79-3 can now be linked and arranged into an apparent time sequence by reference to Figs. 11b and 11d, based on the premise that the poles RR_B, RR_D, RR_E, RR_F and CP_A have been separated by tectonic rotations. This time sequence was recorded completely by the oldest formations, but only the latest events were recorded by the youngest unit considered in this study (lower Coppercap Formation). Directions with the greatest apparent rotations are interpreted as recording the oldest magnetizations; those closest to the P2 cluster, as the youngest. It is possible that the picture is more complex because the magnitude of rotation may vary from site to site and may erratically increase or decrease through time.

The first event recorded paleomagnetically is deposition of detritus, including mostly labile mafic minerals and lesser magnetite and hematite opaques (Lustwerk, 1990) from Katherine and Little Dal groups in grabens next to "active fault scarps" to form rhythmic siltstones (RR3, 4) of the Redstone River Formation (Jefferson, 1983). Blocks in these graben systems had already begun to rotate prior to deposition of Redstone River Formation, as evidenced by the small-circle distribution of paleomagnetic results from the Thundercloud Formation (Fig. 13b).

The block that includes section 79-3 contains evidence, in component RR_E, of continued rotations during deposition of the Redstone River Formation. This component is linked by leaching experiments and petrography to hematite coatings about quartz. It may be either pre-depositional or very early diagenetic, but likely the latter; because magnetic moments associated with predepositional hematite coatings probably would be randomly scattered rather than statistically aligned in the Earth's field. Hematite coatings (RR_E) as well as some of the first phase disseminated hematite (or magnetite) hosting RR_D may have been preserved by quartz overgrowths, as well as by the lack of any reductants in this non-vegetated environment.

During and near the end of rotation of the block, the CP1 member was deposited over siltstones of Redstone River Formation. Carbonate beds and adjacent siltstones within CP1 were mineralized by iron sulphides (not necessarily copper) during very early diagenesis. This could have prevented complete oxidation of detrital magnetite in the adjacent green siltstones. This is suggested by the trace magnetite that is inferred only from the magnetic characteristics of the green siltstones (Fig. 10) immediately beneath Bed 3 (Fig. 2). The green siltstones and adjacent microbial limestones have virtually the same magnetic direction, which agrees well with RR_D from the underlying Redstone River Formation. CP_A has the magnetic characteristics (Fig. 10) of magnetite, which could be a primary detrital phase, but is too fine-grained and rare to be distinguished petrographically. Like the sulphides in these sediments, the magnetite could alternatively be a diagenetic product of the reduction process.

RR_D may, like CP_A, reside in magnetite instead of disseminated hematite, because it has unblocking temperatures to about 580°C, in the magnetite range. RR_D hematite or magnetite could have formed throughout the Redstone River Formation at the same time as microbial processes were locally reducing siltstones adjacent to carbonates in the overlying Coppercap member CP1.

Alteration of detrital magnetite to martite (RR_F) may have been completed with continuation of oxidizing conditions, after or near the

end of rotation. At about the same time, oxidation of the Redstone River Formation as a whole is recorded by component RR_B, which has been traced to hematite pigment in the partly silicified fine-grained matrix. In member CP1, the only sample of red siltstone collected from the Coppercap Formation records component CP_B, which has a similar direction to RR_F and RR_B (Fig. 11). Its magnetic characteristics suggest that it is linked to RR_B.

Addition of copper to early diagenetic iron sulphides (Chartrand et al., 1989) may have begun with, and continued after formation of possible magnetite (CP_A) and hematite pigmentation (RR_B), because the copper–iron sulphides invade diagenetic quartz and feldspars. This inference is weak, because there is no guarantee that the diagenetic quartz in the copper beds is temporally linked to the diagenetic quartz that encloses hematite pigment in the red siltstones. The most pervasive hematite pigment, that which carries CP_B and RR_B, probably was introduced while the block was still rotating, because RR_B appears to be rotated slightly clockwise with respect to reference directions T_B and K_B (Figs. 11a, 11b). The RR_B hematitization event is suggested to be close in time to the main copper–iron–sulphide mineralization, by the small rotational difference between CP_A and RR_B, and by the inference that hematitization had to precede deposition of the sulphidic Coppercap Formation, which overlapped the entire red bed sequence onto unconformably underlying platformal carbonates.

The P2 hematite mineralization in Katherine, Little Dal, and Coates Lake groups is interpreted to have coincided with incipient rifting and the attendant circulation of oxidizing saline groundwaters (Park and Aitken, 1986b). Sediments in the rift zone (Coates Lake Group) and immediately to the east in the Outer Fold Belt (Katherine Group, Tsezotene Formation, lower Little Dal Group) show much evidence for the deep pigmentation and leaching effects of oxidizing fluids that would predictably have been contemporaneous with the complex block movements inferred for the Coates Lake Group during incipient and episodic rifting.

One line of evidence for the deep fluid circulation is the magnetically complex and stratigraphically extensive (down to Tsezotene Formation, T_B) hematite mineralization. This mineralization is paleomagnetically linked with the main hematitization of Coates Lake Group. A second line of evidence is the extent of rutile development. In ordinary sedimentary deposits the opaque alteration product, rutile, is only rarely observed (Ramsdohr, 1969), but in the Mackenzie Mountains region, rutile, which remains when iron is removed from Fe–Ti oxides, is common to abundant at all stages of development. It replaces all detrital hematite to varying degrees and apparently the hematite pigment to a lesser extent. Silica cement seems to postdate the rutile alteration of the detrital opaques, but may predate some of the "rutilized" pigment. Extensive acid leaching in the laboratory duplicates the natural rutilization of all opaques present.

In summary we infer that four magnetizations were preserved in rapid succession, during and at the end of tectonic rotation, and before deposition of the upper members (CP2–CP7) of Coppercap Formation: (1) very early diagenetic hematitizations (RR_E, RR_D); (2) primary detrital or very early diagenetic magnetite phase preserved or formed in reduced carbonates and siltstones (CP_A); (3) martite after detrital magnetite (RR_F); (4) pervasive disseminated hematite pigment (RR_B, CP_B) inferred to be approximately synchronous with copper mobilization out of the red beds and fixing in the microbial carbonate beds.

A fifth diagenetic event, the invasion of chloritic alteration and iron $\pm$ copper sulphides into red siltstones of CP1, does not appear to be associated with a resolvable paleomagnetic component, and is presumed to considerably post-date deposition of CP1 member. Similar chloritic alteration and

TABLE 7
Site coordinates and bedding data

Section 79-3 (126.6°W, 62.7°N)	
1.	259, 45
2.	263, 38
3.	270, 40
Section 79-4 (127,0°W, 63.5°N)	
4.	268, 38
5.	258, 37
6.	264, 39
7.	264, 38
8.	257, 38
Section 79-9 (127.0°W, 63.4°N)	
9.	289, 24
10.	290, 23
11.	284, 25
12.	291, 24
Section 79-10 (126.6°W, 62.5°N)	
13.	244, 60
14.	265, 55
15.	263, 49
Section X (127.5°W, 63.8°N)	
16.	205, 10
17.	205, 10
19.	227, 09
20.	227, 09
Section 79-6 (127.2°W, 63.7°N)	
18.	227, 32
Section 79-6 (126.4°W, 62.4°N)	
21.	052, 65
22.	063, 65
23.	060, 60

Values in degrees. First number is down-dip azimuth; second is dip.

bleaching of the basal Sayunei siltstones is located at their basal contact with grey Coppercap limestones. The above paleomagnetic events provide a positive test of previously proposed models for the formation of the Redstone copper belt.

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